

第三章 模拟调制系统

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网络与信息安全学院

调制的概念

- 一、调制的目的
- 二、调制的定义
- 三、调制的分类
- 四、调制的模型

调制的概念

一、调制的目的

- 将调制信号转换成**适合**于信道传输的已调信号
【例】 3kHz信号直接耦合到天线，形成有效辐射的天线尺寸要达到**25km**，调制到900MHz，天线长度仅需**8cm**
- 通过把多个基带信号分别搬移到不同的载频处，实现信道的**多路复用**，提高信道利用率
- 扩展信号带宽，提高系统抗干扰、抗衰落能力
- 实现传输带宽与信噪比之间的互换

调制的概念

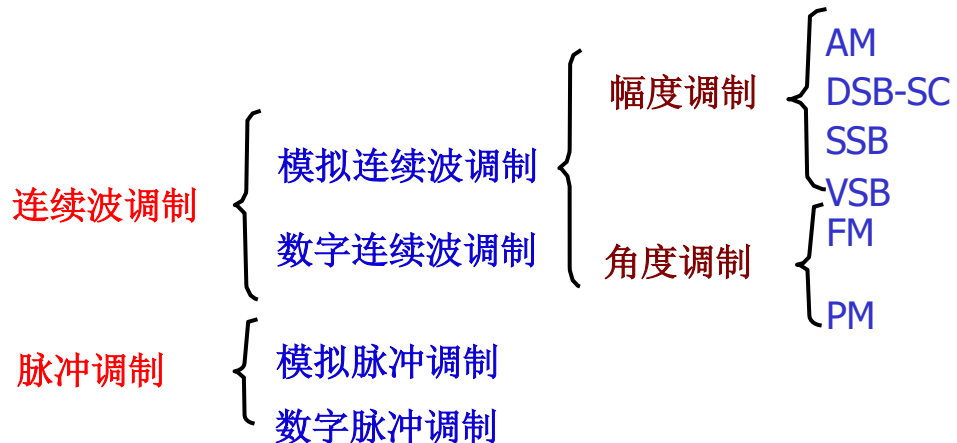
二、调制的定义

- 调制就是把信号转换成适合在信道中传输形式的过程
- 广义调制分为基带调制和载波调制，载波调制是用调制信号（基带信号）控制载波的某个（或几个）参数的过程
- 狭义的调制就是指载波调制

调制的概念

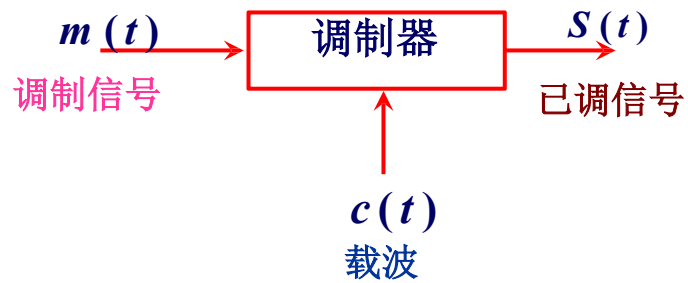
三、调制的分类

- 根据载波类型、调制信号、载波参数变化的不同进行分类



调制的概念

四、调制模型



幅度调制——用消息去控制正弦载波的**幅度**变化

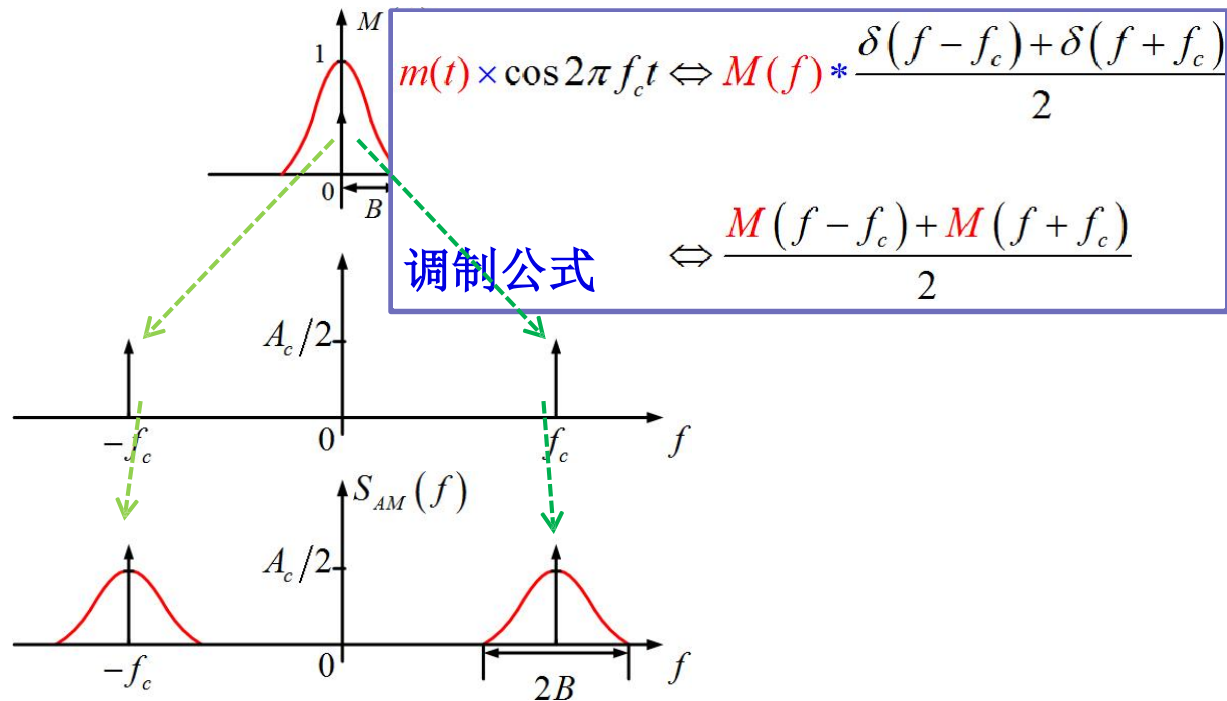
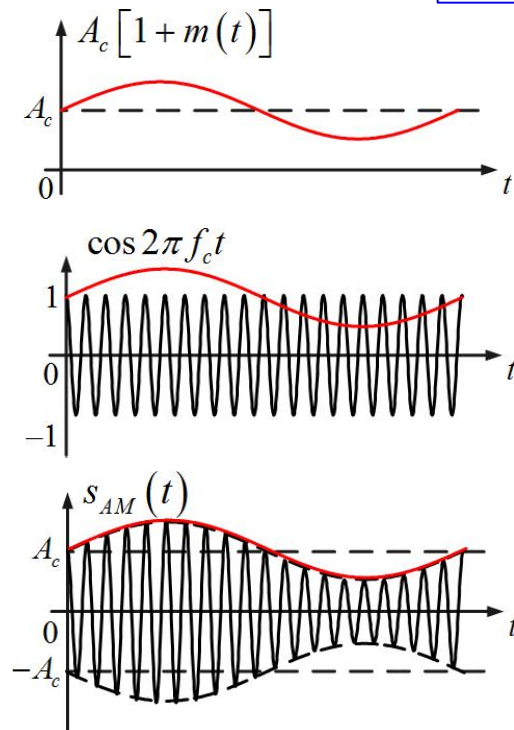
- ✓ **常规调幅 (Conv. AM)**
- ✓ 抑制载波双边带调幅 (DSB-SC)
- ✓ 单边带调幅 (SSB)
- ✓ 残留边带调幅 (VSB)

第1部分 常规调幅

常规调幅 (conventional AM)

1. 基本原理

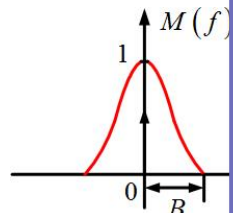
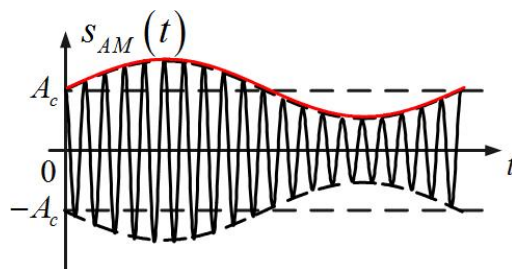
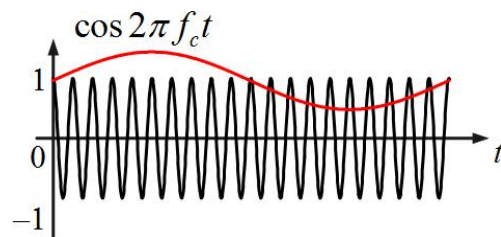
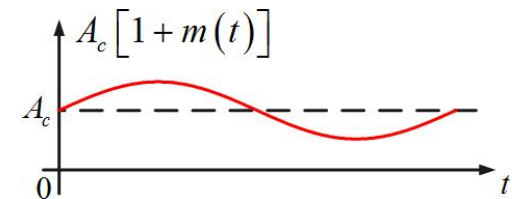
$$s_{AM}(t) = A_c [1 + m(t)] \cos 2\pi f_c t$$



常规调幅 (conventional AM)

1. 基本原理

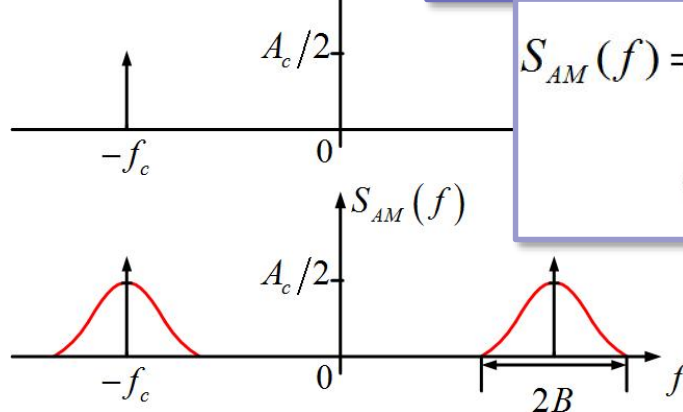
$$s_{AM}(t) = A_c [1 + m(t)] \cos 2\pi f_c t$$



$$m(t) \times \cos 2\pi f_c t \Leftrightarrow M(f) * \frac{\delta(f-f_c) + \delta(f+f_c)}{2} + 1 + \delta(f)$$

$$\Leftrightarrow \frac{M(f-f_c) + M(f+f_c)}{2}$$

调制公式



$$S_{AM}(f) = \frac{A_c}{2} [M(f-f_c) + M(f+f_c)] + \frac{A_c}{2} [\delta(f-f_c) + \delta(f+f_c)]$$

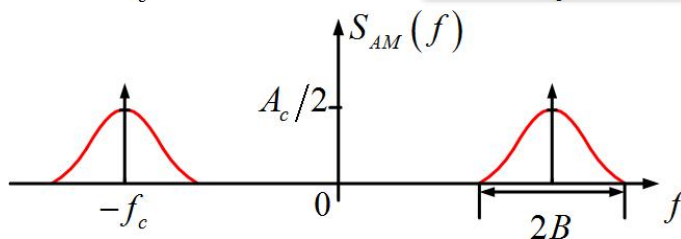
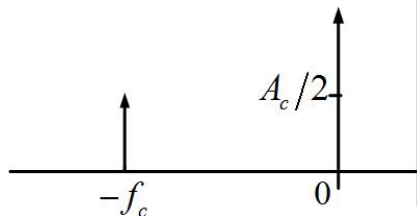
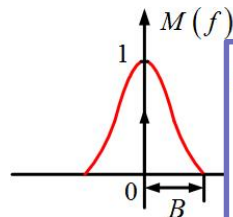
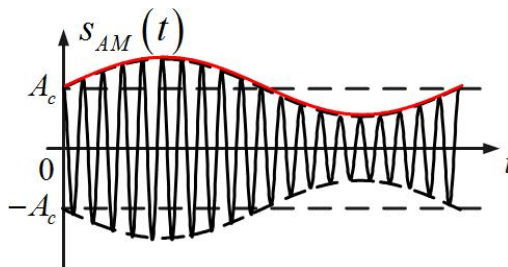
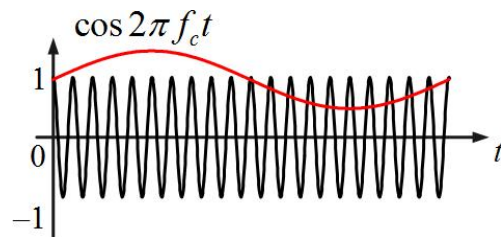
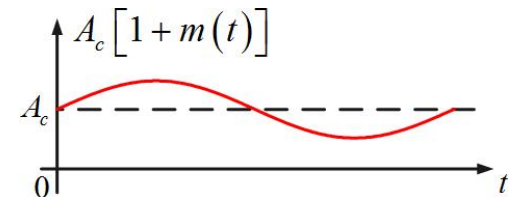
常规调幅 (conventional AM)

1. 基本原理

$$s_{AM}(t) = A_c [1 + m(t)] \cos 2\pi f_c t$$

频谱特点

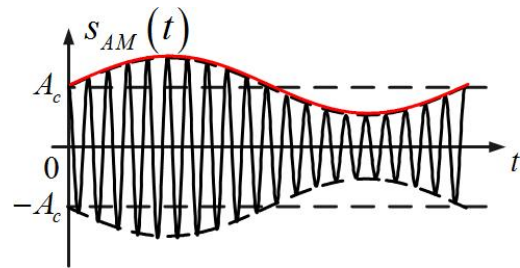
- 1) 频带信号：位于载频 f_c ，带宽 $B_T = 2B$
- 2) 上、下两个边带：
上边带 (USB) —— 高于 f_c 的边带
下边带 (LSB) —— 低于 f_c 的边带
- 3) $\pm f_c$ 处有两个冲激，有纯载波。



常规调幅 (conventional AM)

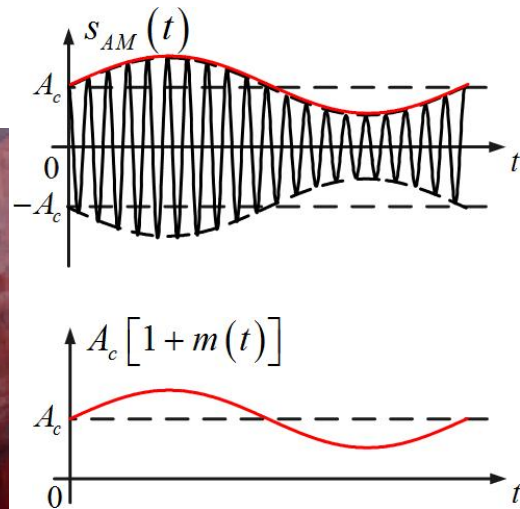
2. 波形特点:

消息突显在载波包络上



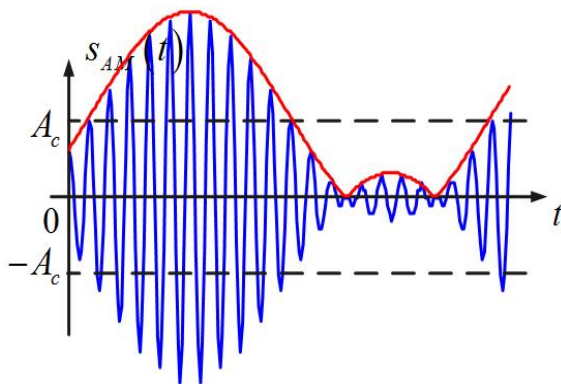
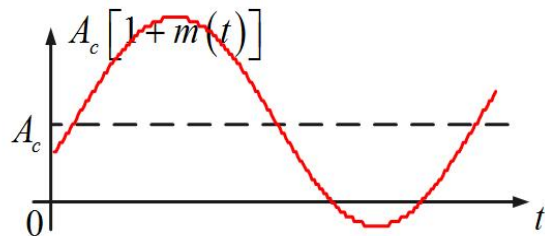
常规调幅 (conventional AM)

2. 波形特点：消息突显在载波包络上



常规调幅 (conventional AM)

2. 波形特点：消息突显在载波包络上



定义调幅指数：

$$\beta_{AM} = \frac{\max[s_{AM}(t)] - A_c}{A_c}$$
$$= \max|m(t)|$$

$\beta_{AM} \leq 1$ ：正常情况

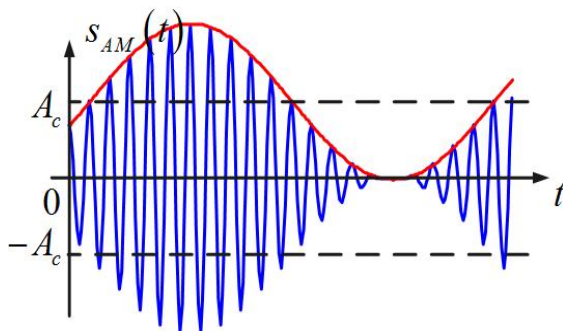
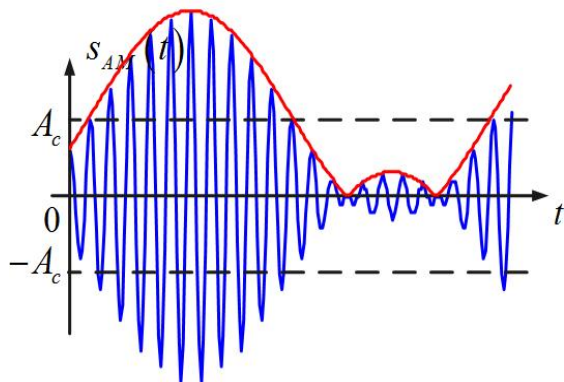
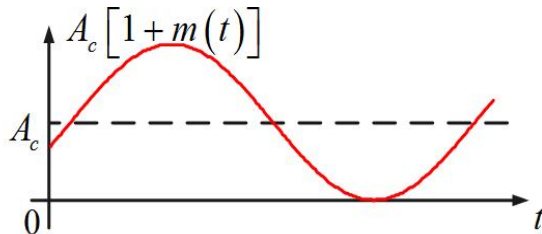
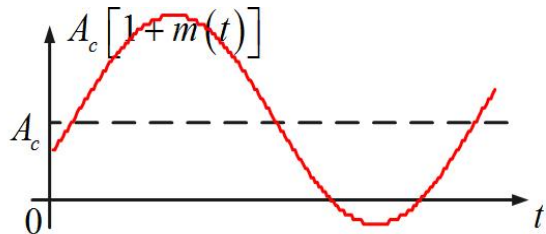
$\beta_{AM} = 1$ ：临界调制

$\beta_{AM} > 1$ ：过调制

$$A_c [1 + m(t)] \cos 2\pi f_c t$$

常规调幅 (conventional AM)

2. 波形特点：消息突显在载波包络上



定义调幅指数：

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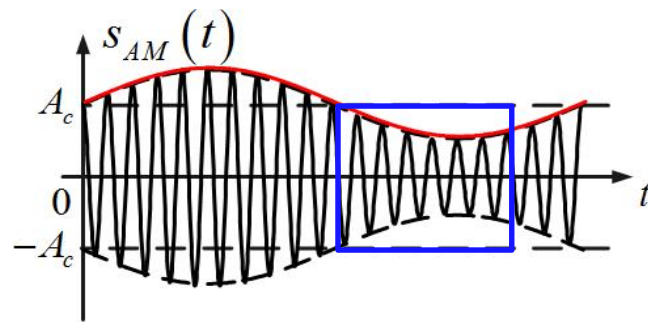
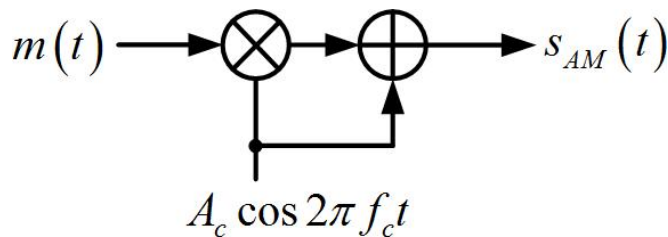
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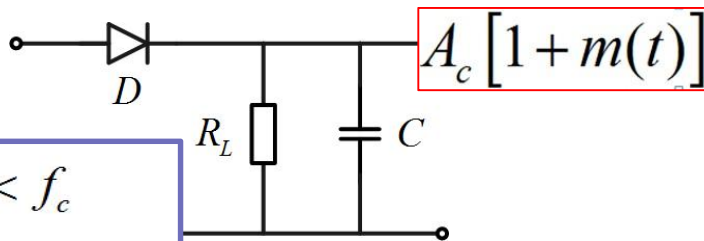
常规调幅 (conventional AM)

3. 发送与接收 $A_c [1 + m(t)] \cos 2\pi f_c t$

产生方案



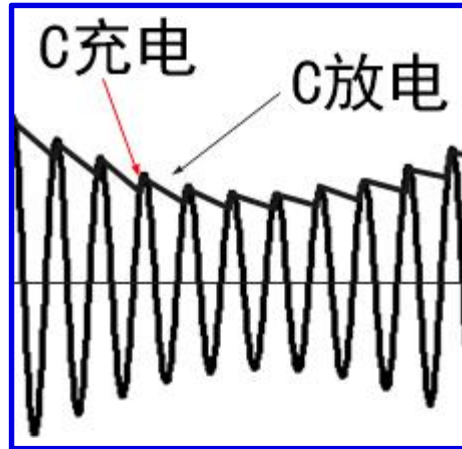
包络检波器



$$B \ll \frac{1}{2\pi R_L C} \ll f_c$$

4000

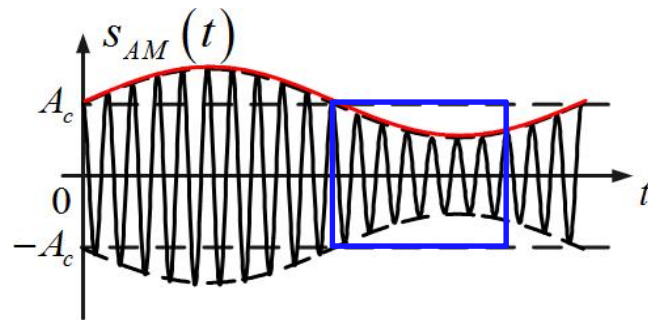
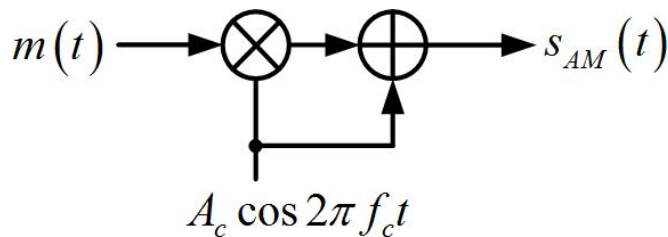
500kHz



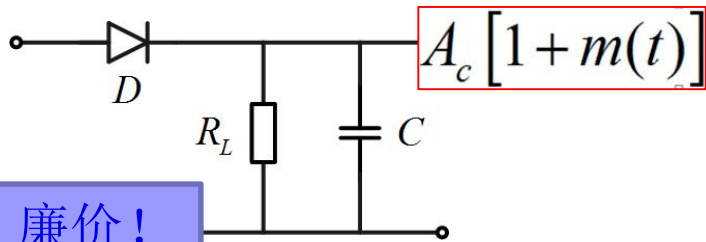
常规调幅 (conventional AM)

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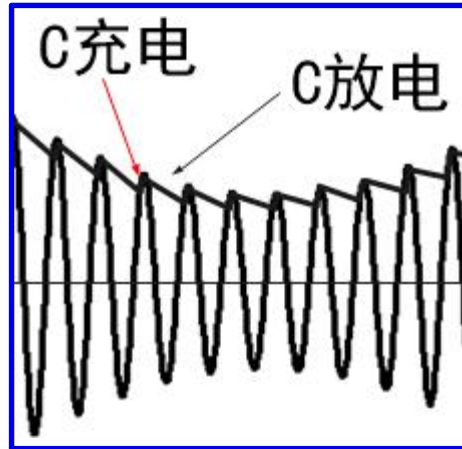
产生方案



包络检波器



非常简单、廉价！



常规调幅 (conventional AM)

4. 功率与效率 $A_c [1 + m(t)] \cos 2\pi f_c t$

$$P_{AM} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T s_{AM}^2(t) dt = \overline{s_{AM}^2(t)} = \overline{A_c^2 [1 + m(t)]^2 \cos^2 2\pi f_c t}$$
$$= \frac{A_c^2}{2} + \frac{A_c^2}{2} \overline{m^2(t)} = P_c + P_m$$

$$A_c \cos 2\pi f_c t$$

$$A_c m(t) \cos 2\pi f_c t$$

常规调幅 (conventional AM)

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$$= \frac{A_c^2}{2} + \frac{A_c^2}{2} \overline{m^2(t)} = P_c + P_m$$

$$PAPR_{m(t)} = \frac{\max |m(t)|^2}{\overline{m^2(t)}}$$

定义调制效率

$$\eta_{AM} = \frac{P_m}{P_{AM}} = \frac{P_m}{P_m + P_c} = \frac{\overline{m^2(t)}}{1 + \overline{m^2(t)}}$$

$$\eta_{AM} = \frac{1}{1/\overline{m^2(t)} + 1}$$

常规调幅 (conventional AM)

4. 功率与效率 $A_c [1 + m(t)] \cos 2\pi f_c t$

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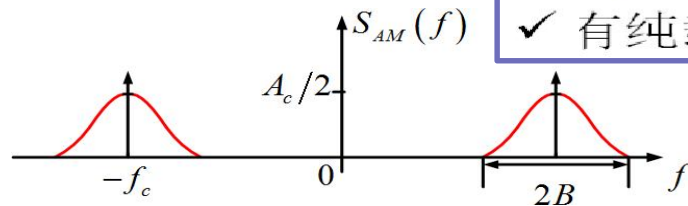
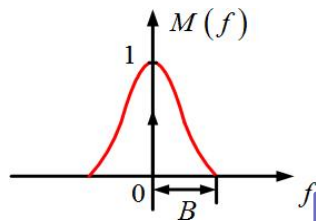
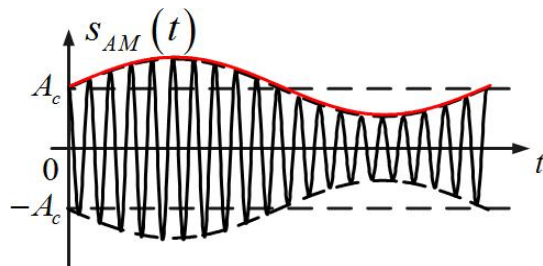
$$\eta_{AM_临界} = \frac{1}{1 + PAPR_{m(t)}}$$

例如，正弦测试信号 $PAPR=2$ ， $\eta_{AM} = 33\%$ 。语音更低

常规调幅 (conventional AM)

小结:

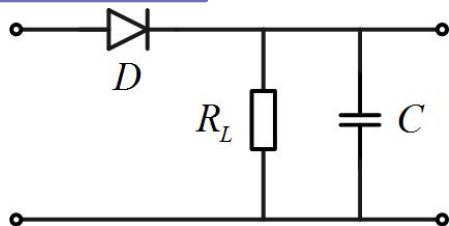
$$s_{AM}(t) = A_c [1 + m(t)] \cos 2\pi f_c t$$



✓ $B_T = 2B$
✓ 有纯载波

$$\beta_{AM} = \max |m(t)| \leq 1$$

包络检波器



接收简单
廉价实用

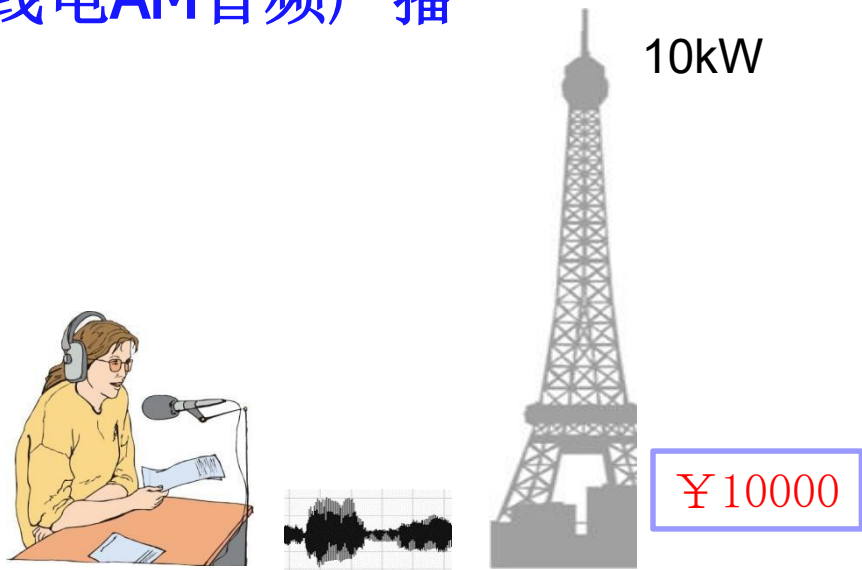
效率很低

$$\eta_{AM} = \frac{\overline{m^2(t)}}{1 + \overline{m^2(t)}} = \left[1 + PAPR_{m(t)} \right]^{-1} \Big|_{\text{临界}}$$

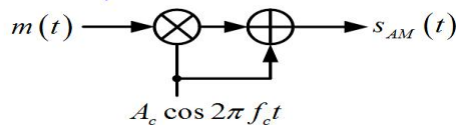
正弦测试信号: $\eta_{AM} = 33\%$

常规调幅 (conventional AM)

无线电AM音频广播



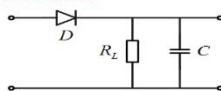
产生方案



$$¥10,000 + 1000,000 \times ¥1 = ?$$



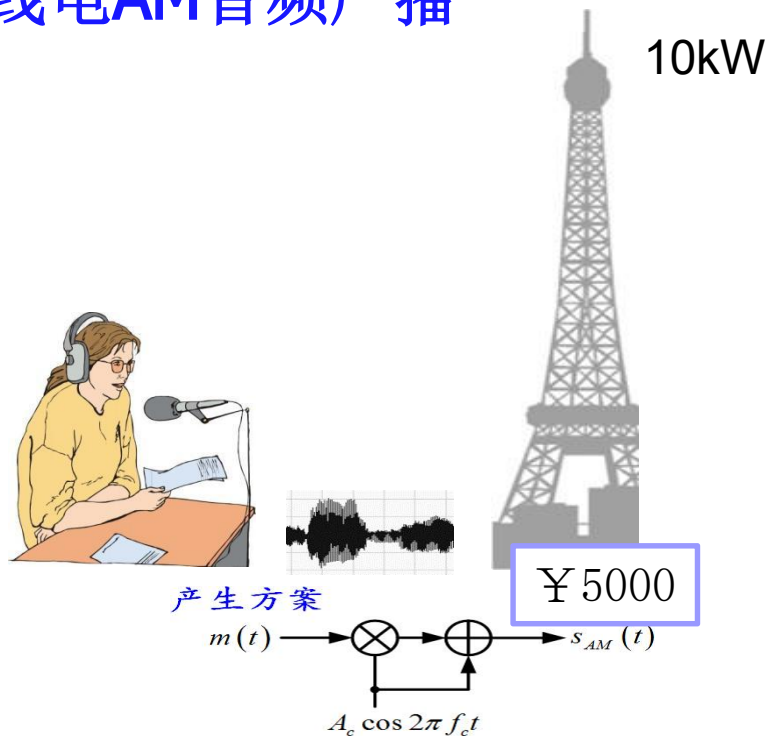
包络检波器



非常简单实用!

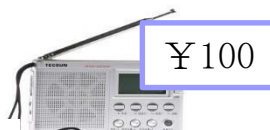
常规调幅 (conventional AM)

无线电AM音频广播

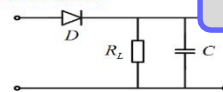


$$¥10,000 + 1,000,000 \times ¥1 = ?$$

$$¥5,000 + 1,000,000 \times ¥100 = ?$$



包络检波器



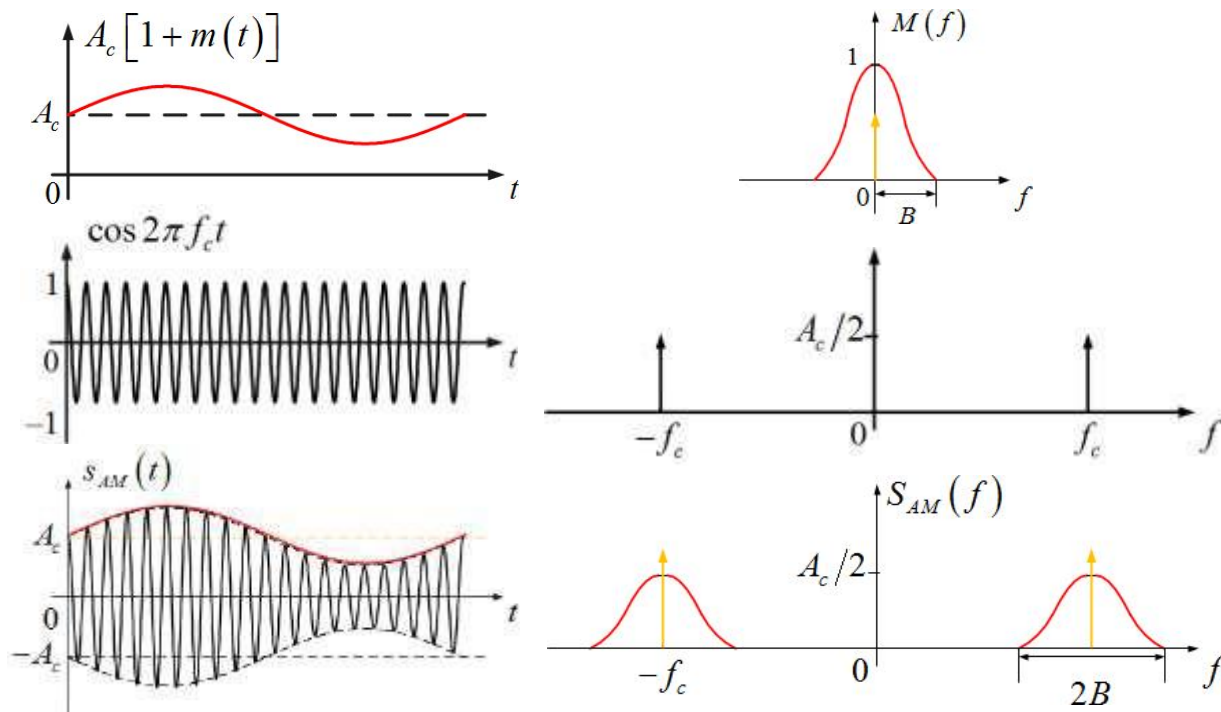
非常简单实用!

第2部分 抑制载波双边带调幅

抑制载波双边带调幅 (DSB-SC)

1. 基本原理

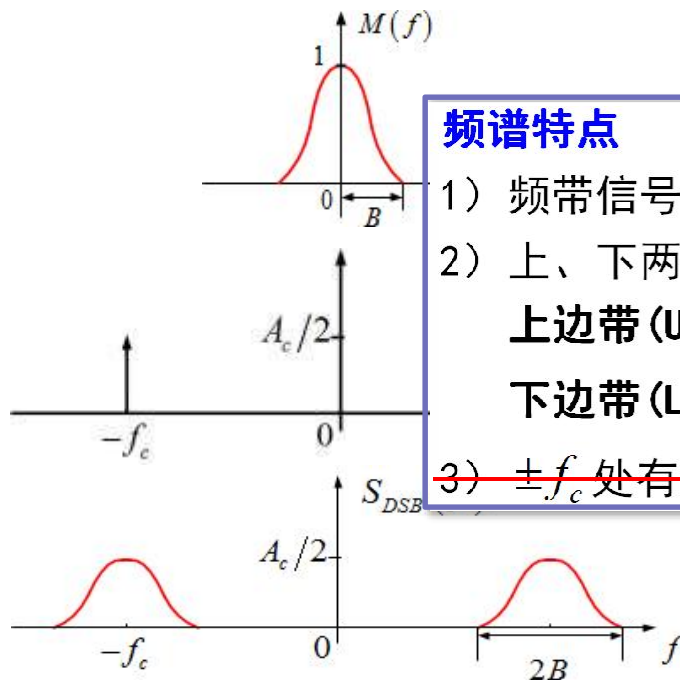
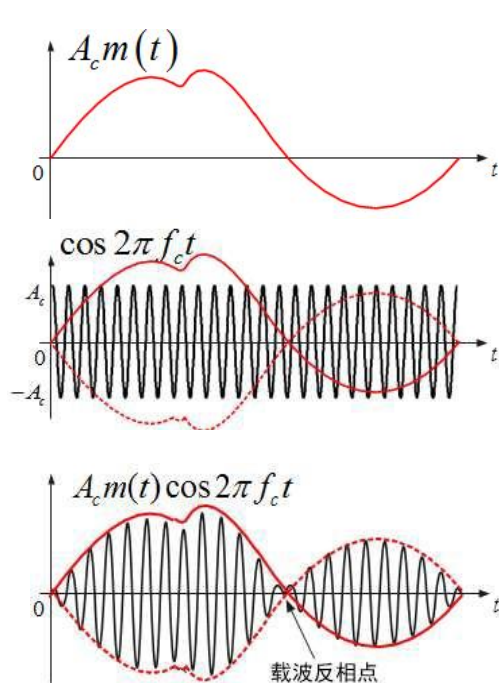
$$s_{AM}(t) = A_c [1 + m(t)] \cos 2\pi f_c t$$



抑制载波双边带调幅 (DSB-SC)

1. 基本原理

$$s_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$$



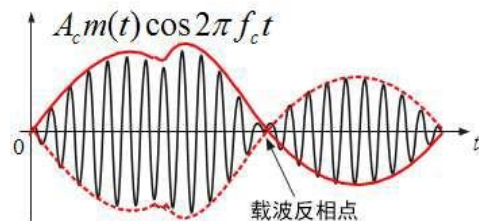
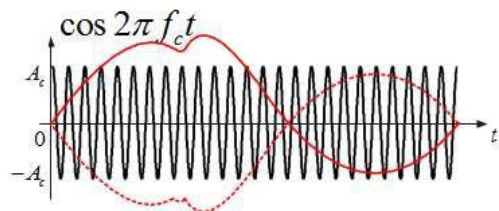
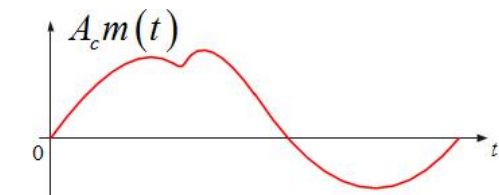
频谱特点

- 1) 频带信号：位于载频 f_c ，带宽 $B_T = 2B$
- 2) 上、下两个边带：
上边带 (USB) —— 高于 f_c 的边带
下边带 (LSB) —— 低于 f_c 的边带
- 3) ~~$\pm f_c$ 处有两个冲激，有纯载波。~~

抑制载波双边带调幅 (DSB-SC)

1. 基本原理

$$s_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$$



功率

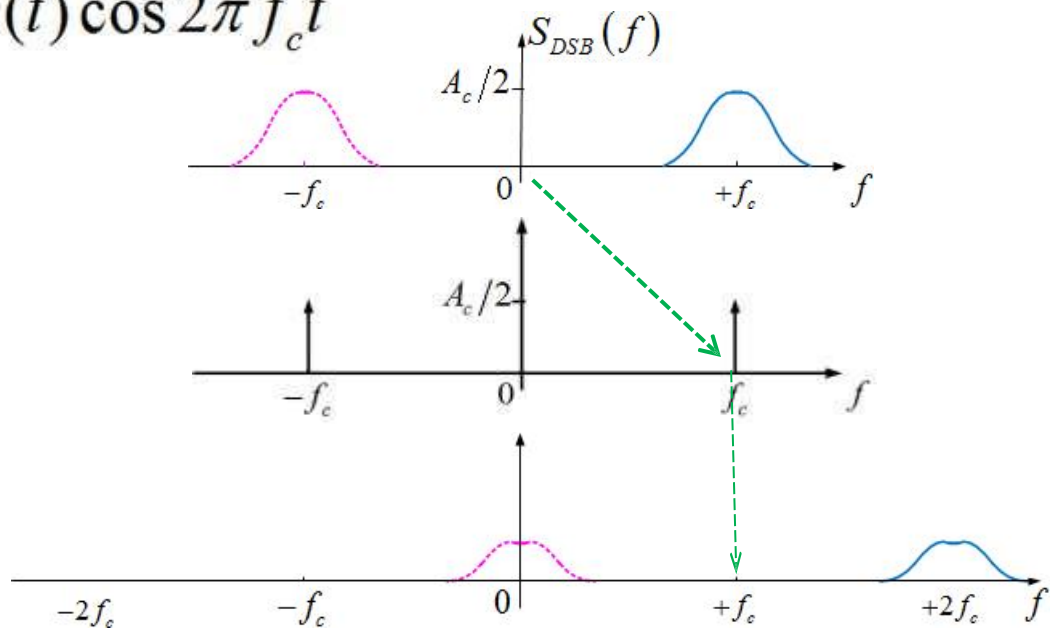
$$\begin{aligned} P_{DSB} &= \overline{s_{DSB}^2(t)} = \overline{A_c^2 m^2(t) \cos^2 2\pi f_c t} \\ &= P_m = \overline{A_c^2 m^2(t)} / 2 \end{aligned}$$

调制效率 $\eta_{DSB} = 1$

抑制载波双边带调幅 (DSB-SC)

2. 接收方法 $s_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$

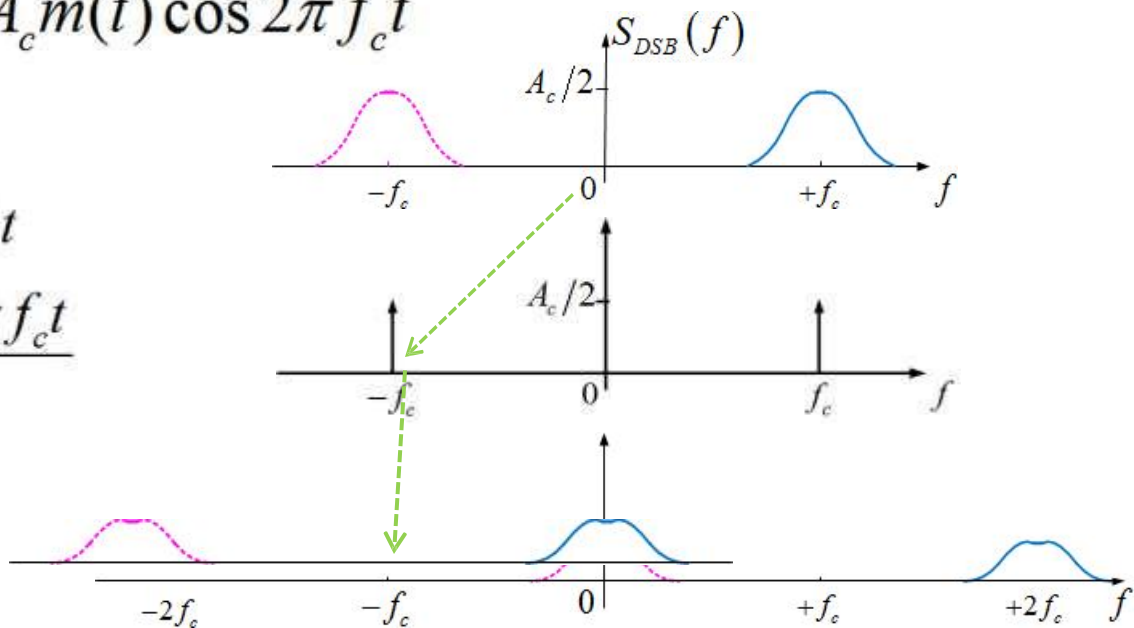
$$\begin{aligned} s_{DSB}(t) \times \cos 2\pi f_c t \\ &= A_c m(t) \cos^2 2\pi f_c t \\ &= A_c m(t) \frac{1 + \cos 4\pi f_c t}{2} \end{aligned}$$



抑制载波双边带调幅 (DSB-SC)

2. 接收方法 $s_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$

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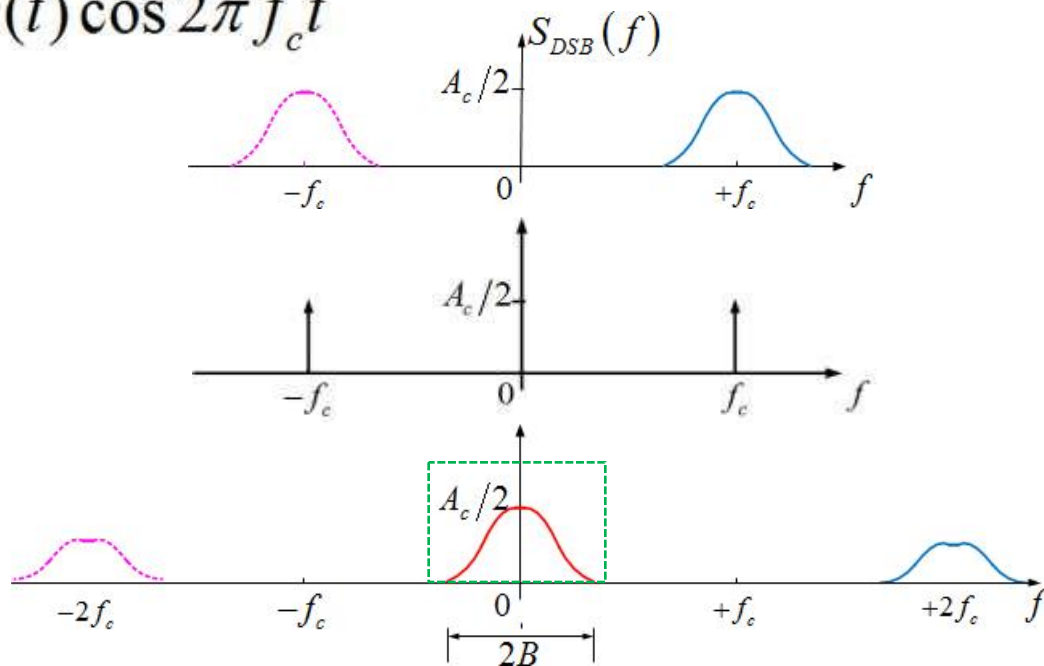


抑制载波双边带调幅 (DSB-SC)

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$$\begin{aligned} LPF_B \{s_{DSB}(t) \cos 2\pi f_c t\} \\ = \frac{1}{2} A_c m(t) \end{aligned}$$

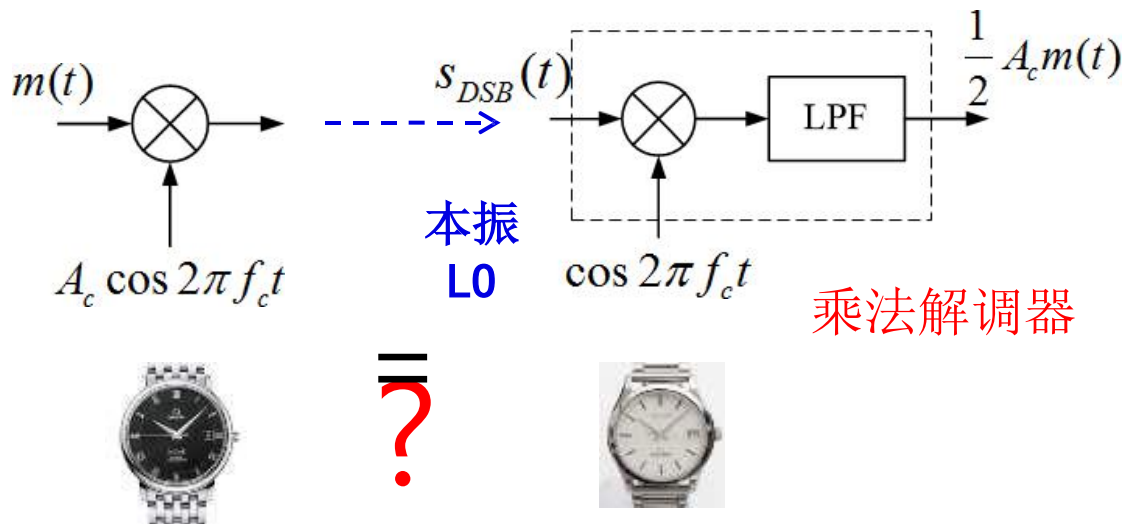


抑制载波双边带调幅 (DSB-SC)

2. 接收方法

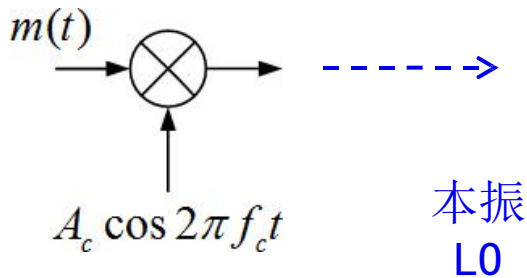
$$\begin{aligned} s_{DSB}(t) \times \cos 2\pi f_c t \\ &= A_c m(t) \cos^2 2\pi f_c t \\ &= A_c m(t) \frac{1 + \cos 4\pi f_c t}{2} \end{aligned}$$

$$\begin{aligned} LPF_B \{ s_{DSB}(t) \cos 2\pi f_c t \} \\ &= \frac{1}{2} A_c m(t) \end{aligned}$$



抑制载波双边带调幅 (DSB-SC)

3. 同步问题

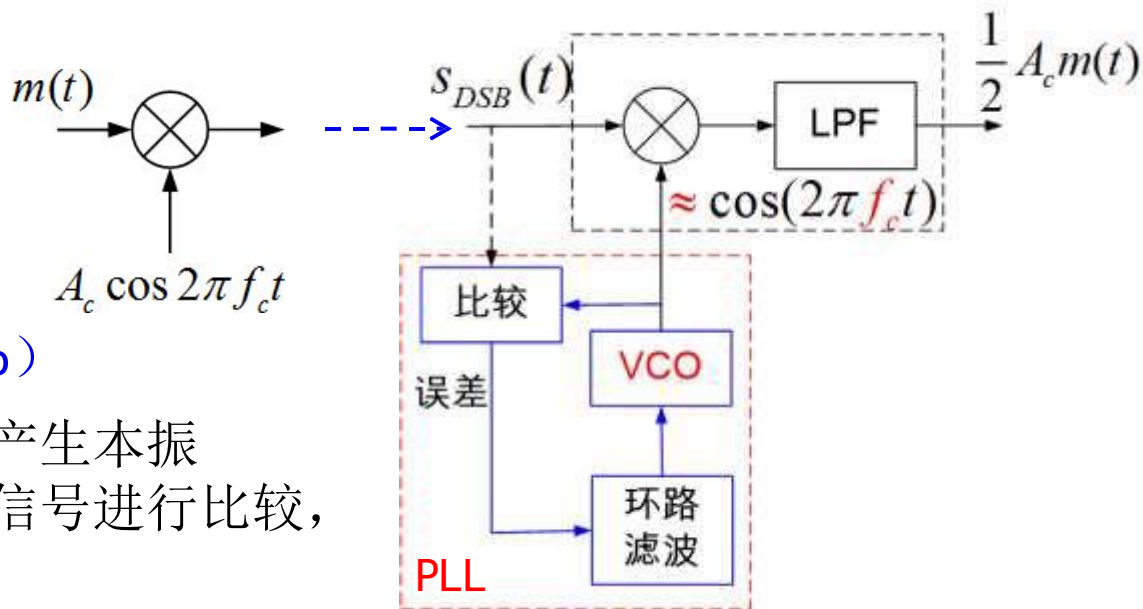


$$\begin{aligned} & LPF_B \{ A_c m(t) \cos 2\pi f_c t \times \cos(2\pi f_o t + \theta_o) \} \\ &= LPF_B \left\{ \frac{1}{2} A_c m(t) \left[\cos[2\pi(f_o - f_c)t + \Delta\theta] + \cos[2\pi(f_o + f_c)t + \Delta\theta] \right] \right\} \\ &= \frac{1}{2} A_c m(t) \cos[2\pi(f_o - f_c)t + \Delta\theta] \end{aligned}$$

两个振荡必须——
频率相同、相位相同
这种条件称为——
同步（或相干）

抑制载波双边带调幅 (DSB-SC)

3. 同步问题



锁相环 (Phase-Lock-Loop)

VCO: 用一个可控振荡器产生本振

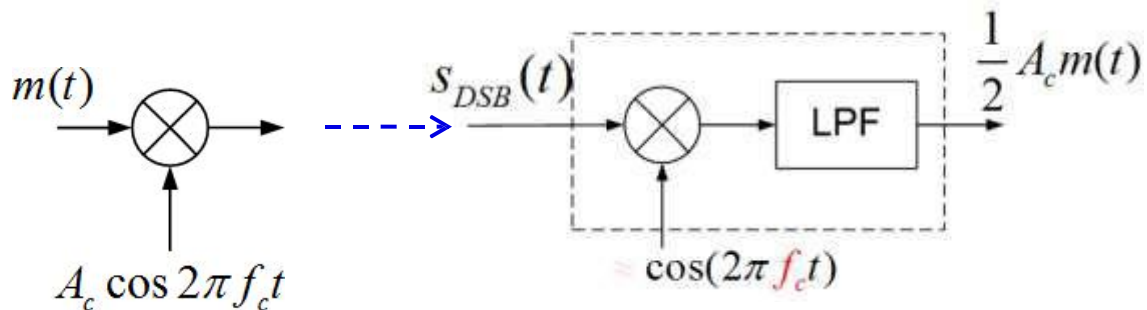
比较: 与含有载波信息的信号进行比较, 看差距多大

反馈: 用误差改进振荡

滤波: “长期观察”, 不要草率调节

抑制载波双边带调幅 (DSB-SC)

3. 同步问题



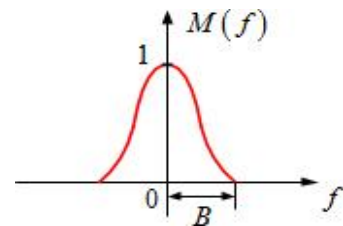
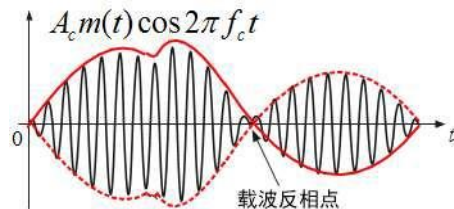
相干解调——需要与载波同步的本地振荡

非相干解调——不需要与载波同步的本地振荡，如，包络检波

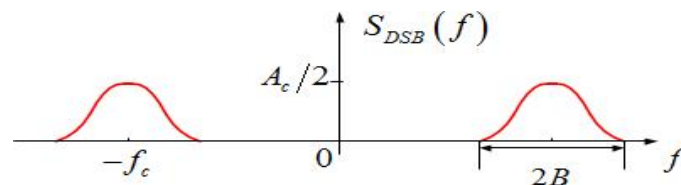
抑制载波双边带调幅 (DSB-SC)

小结:

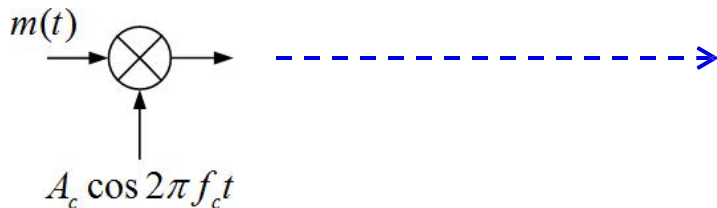
$$s_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$$



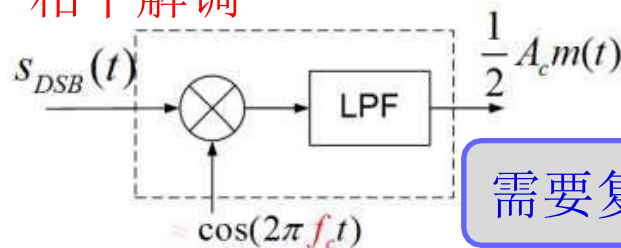
✓ $B_T = 2B$
~~✓ 有纯载波~~



效率高，噪声性能好



相干解调

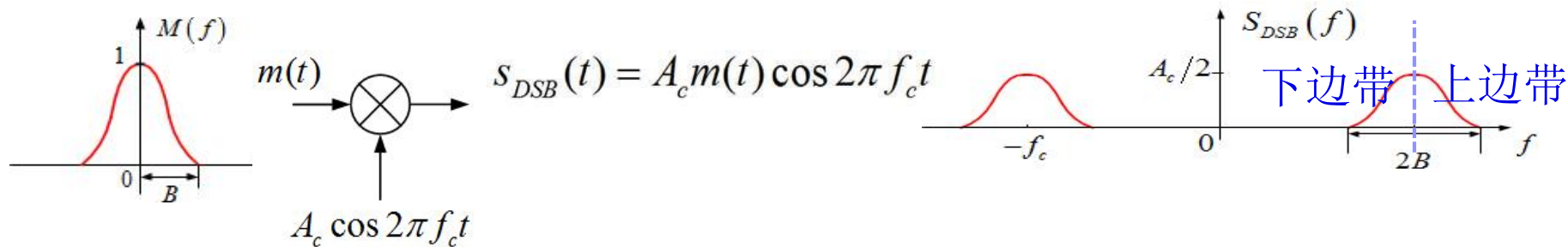


需要复杂的PLL

第3部分 单边带与残留边带调幅

单边带调幅 (SSB)

1. 基本原理



FT性质: $s^*(t) \Leftrightarrow S^*(-f)$

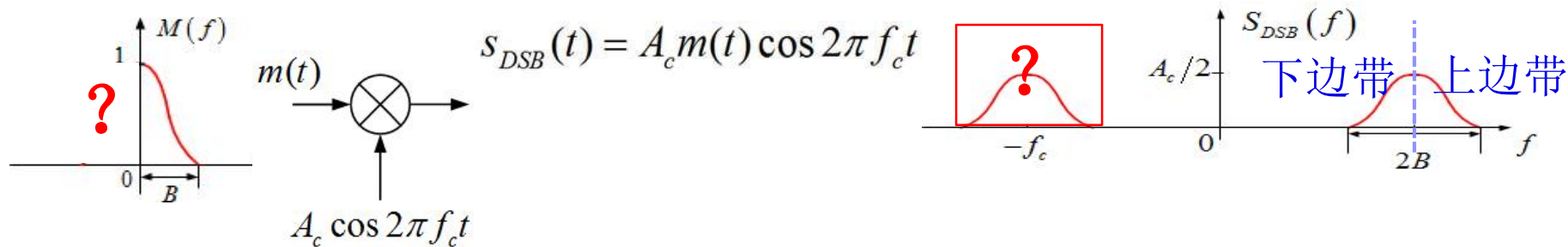
对于实信号: $S(f) = S^*(-f)$, 共轭对称

$$S_R(f) + jS_I(f) = S_R(-f) - jS_I(-f)$$

对称是一种“美”
也包含——冗余

单边带调幅 (SSB)

1. 基本原理



FT性质: $s^*(t) \Leftrightarrow S^*(-f)$

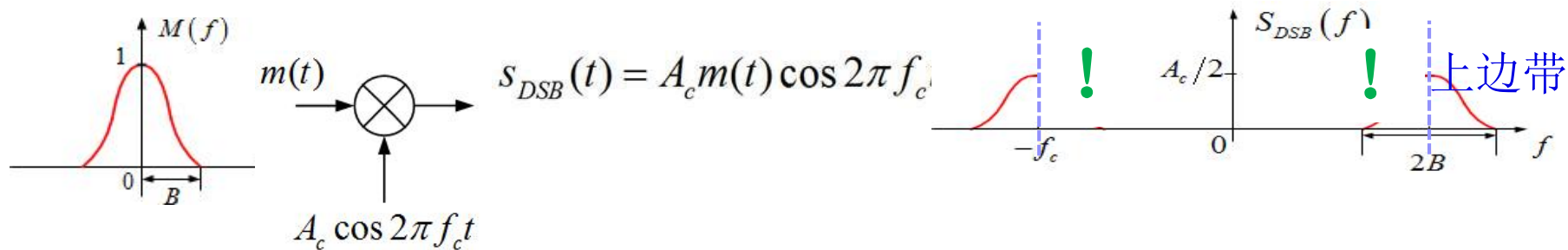
对于实信号: $S(f) = S^*(-f)$, 共轭对称

$$S_R(f) + jS_I(f) = S_R(-f) - jS_I(-f)$$

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单边带调幅 (SSB)

1. 基本原理



FT性质: $s^*(t) \Leftrightarrow S^*(-f)$

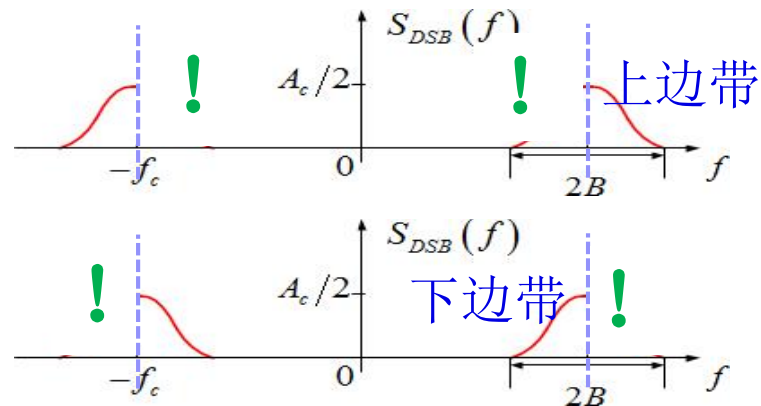
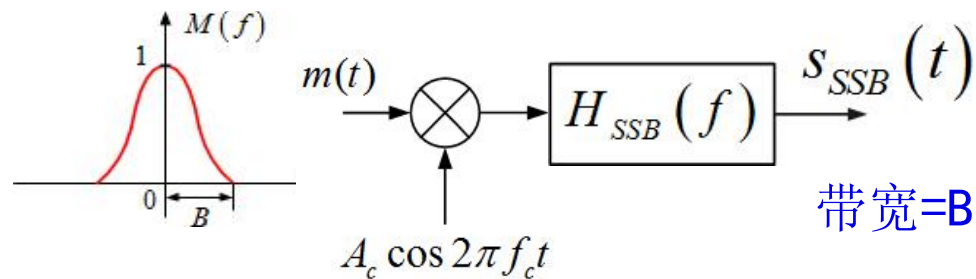
对于实信号: $S(f) = S^*(-f)$, 共轭对称

$$S_R(f) + jS_I(f) = S_R(-f) - jS_I(-f)$$

对称是一种“美”
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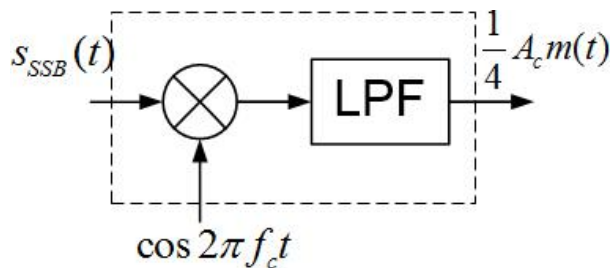
单边带调幅 (SSB)

1. 基本原理

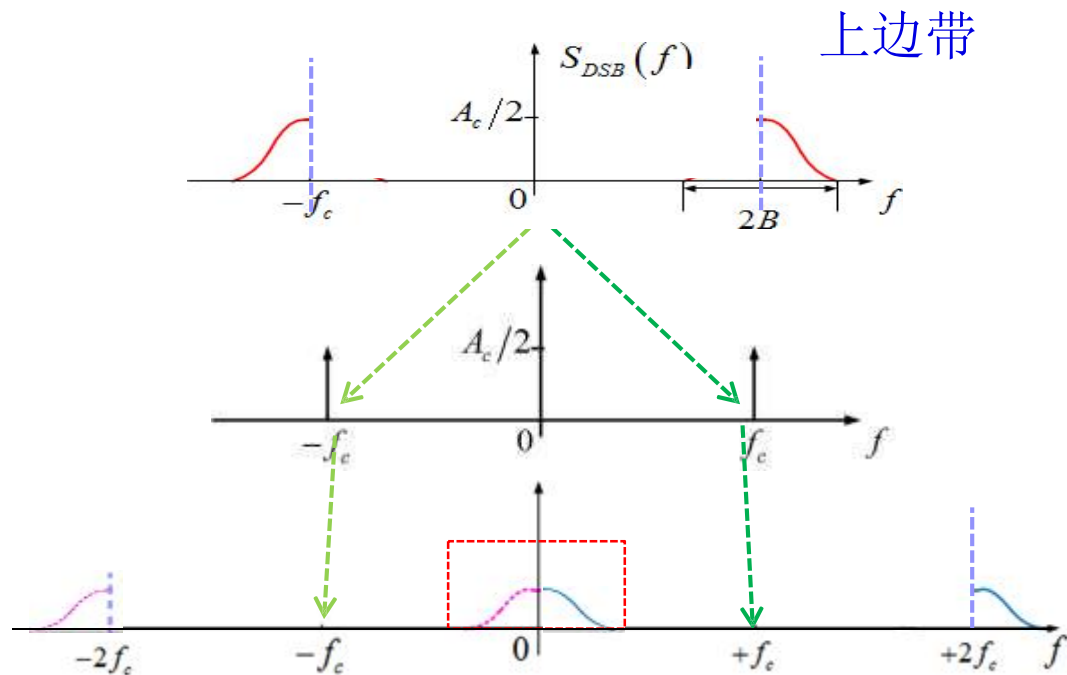


单边带调幅 (SSB)

2. 接收方法



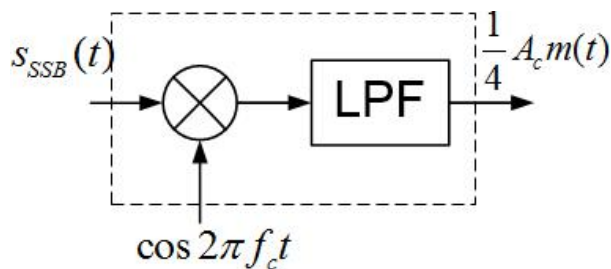
还是相干解调器



单边带调幅 (SSB)

2. 接收方法

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t - \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$



还是相干解调器

$$LPF_B \{ s_{SSB}(t) \times \cos 2\pi f_c t \}$$

$$= LPF_B \left\{ \frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \times \cos 2\pi f_c t \right\}$$

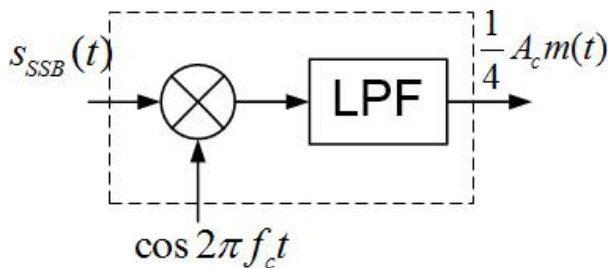
$$= \dots$$

$$= \dots$$

$$= \frac{1}{4} A_c m(t)$$

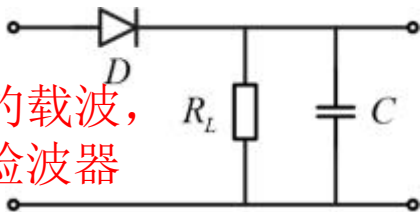
单边带调幅 (SSB)

2. 接收方法



还是相干解调器

如加入足够的载波，
也可用包络检波器



$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos(\alpha + ?)$$

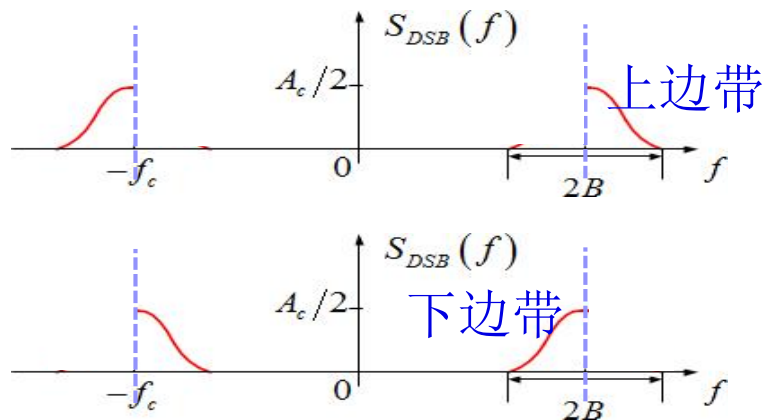
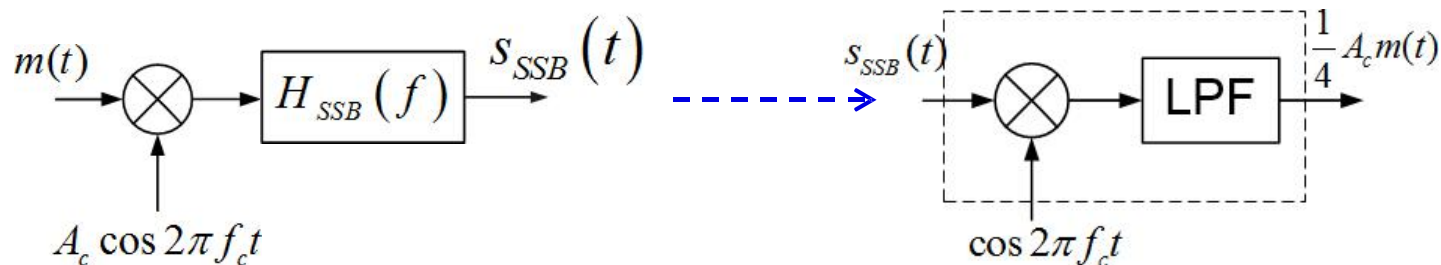
$$\text{包络} \left\{ \frac{1}{2} A_c \begin{bmatrix} [1 + m(t)] \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \right\}$$

$$= \frac{1}{2} A_c \sqrt{[1 + m(t)]^2 + ??^2}$$

$$\approx \frac{1}{2} A_c [1 + m(t)]$$

单边带调幅 (SSB)

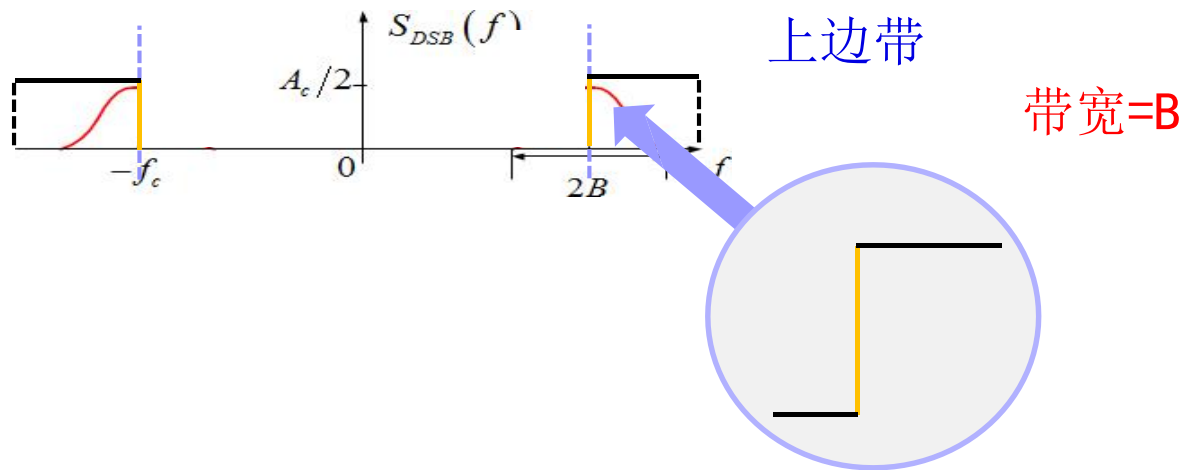
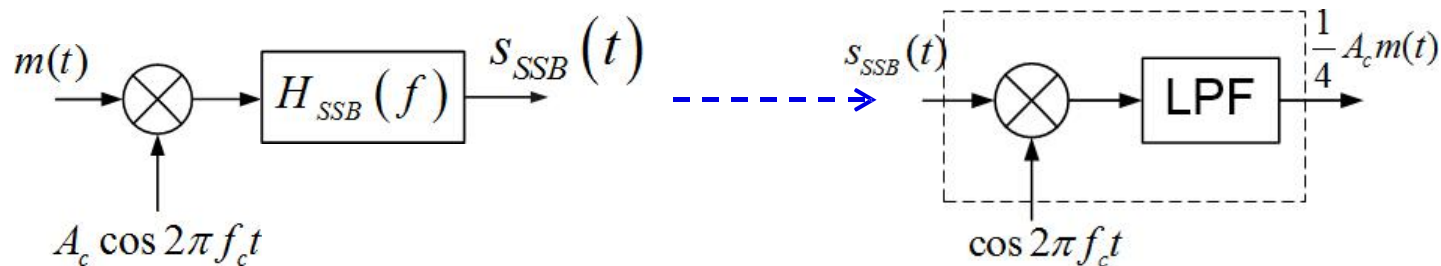
小结:



带宽=B

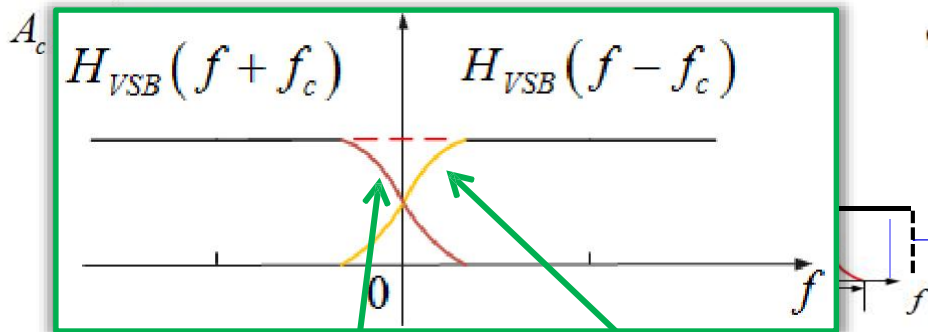
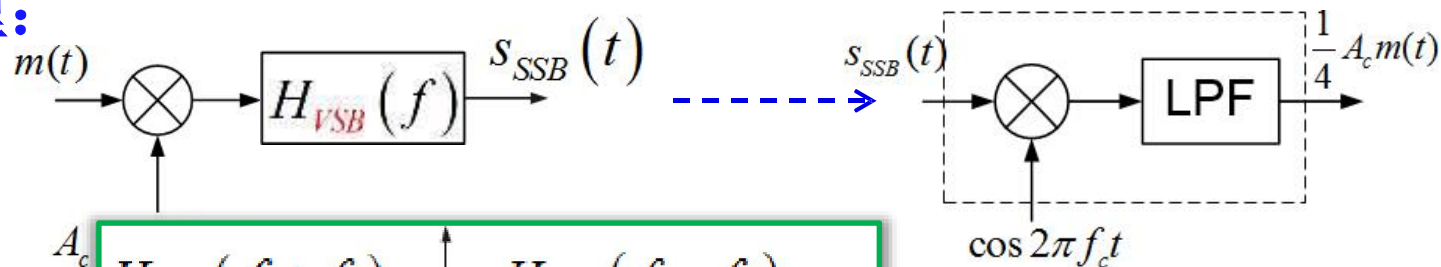
单边带调幅 (SSB)

困难:



残留边带调幅 (VSB)

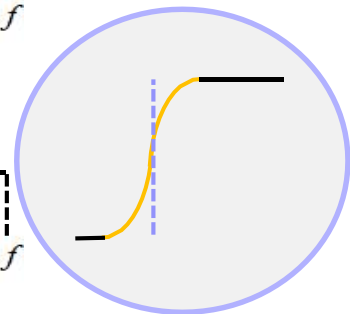
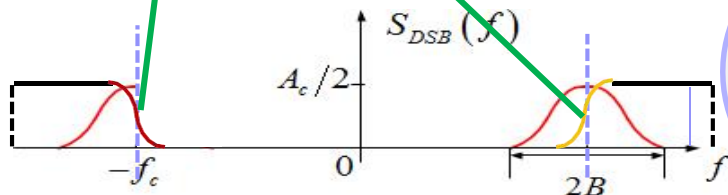
基本原理:



上边带

带宽=B

“残留”一点，
只要过渡带形状
满足对称互补

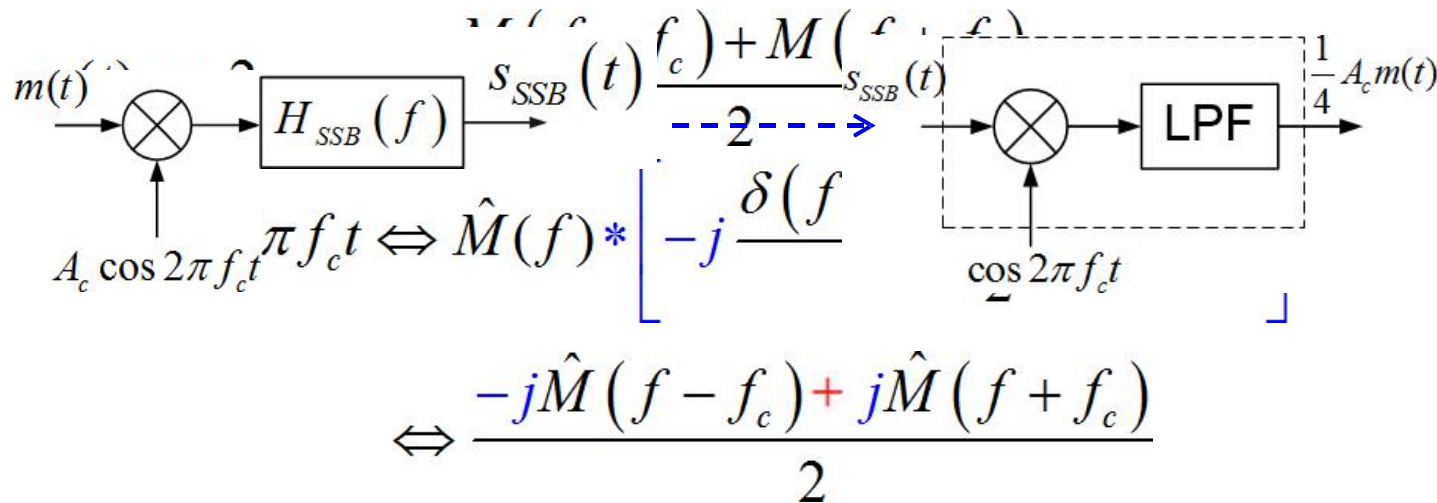


带宽=B+

单边带调幅的相移方法

1. SSB信号公式

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t - \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$



单边带调幅的相移方法

1. SSB信号公式

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t - \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$

$$m(t) \cos 2\pi f_c t \Leftrightarrow \frac{M(f - f_c) + M(f + f_c)}{2}$$

$$\hat{m}(t) \sin 2\pi f_c t \Leftrightarrow \frac{-j\hat{M}(f - f_c) + j\hat{M}(f + f_c)}{2}$$

$$\frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \Leftrightarrow \frac{1}{2} A_c \begin{bmatrix} \frac{M(f - f_c) + M(f + f_c)}{2} \\ \frac{-j\hat{M}(f - f_c) + j\hat{M}(f + f_c)}{2} \end{bmatrix}$$

单边带调幅的相移方法

1. SSB信号公式

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t - \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$

$$\frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \Leftrightarrow$$

$$\frac{1}{2} A_c \begin{bmatrix} \frac{M(f - f_c) + M(f + f_c)}{2} \\ \frac{+j\hat{M}(f - f_c) - j\hat{M}(f + f_c)}{2} \end{bmatrix}$$

$$\hat{m}(t) = m(t) * \frac{1}{\pi t}$$

$$\hat{M}(f) = M(f) \times [-j \operatorname{sgn}(f)]$$

$$j\hat{M}(f) = -j^2 M(f) \operatorname{sgn}(f)$$

$$= \begin{cases} +M(f), & f > 0 \\ -M(f), & f < 0 \end{cases}$$

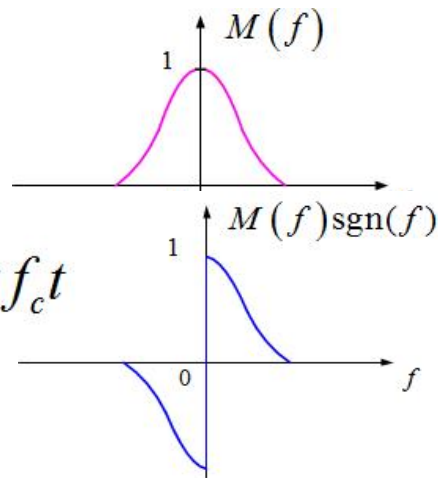
单边带调幅的相移方法

1. SSB信号公式

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$$\frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \Leftrightarrow$$

$$\frac{1}{2} A_c \begin{bmatrix} \frac{M(f-f_c) + M(f+f_c)}{2} \\ \frac{+j\hat{M}(f-f_c) - j\hat{M}(f+f_c)}{2} \end{bmatrix}$$



$$\begin{aligned} j\hat{M}(f) &= -j^2 M(f) \operatorname{sgn}(f) \\ &= \begin{cases} +M(f), & f > 0 \\ -M(f), & f < 0 \end{cases} \end{aligned}$$

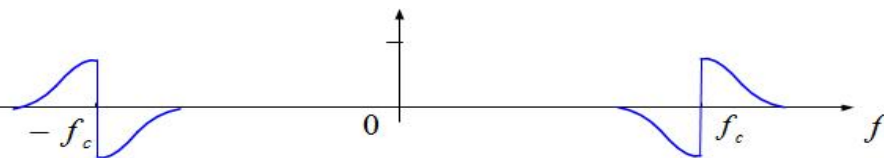
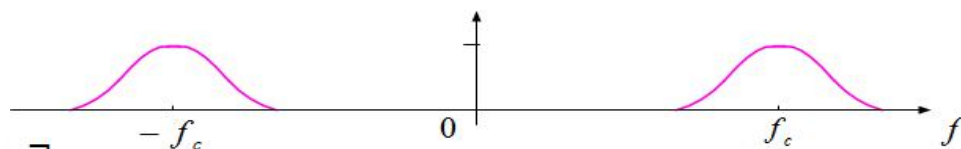
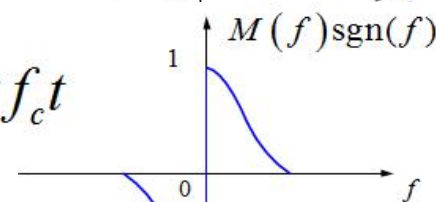
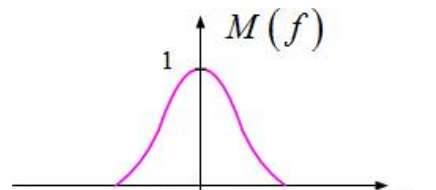
单边带调幅的相移方法

1. SSB信号公式

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t - \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$

$$\frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \Leftrightarrow$$

$$\frac{1}{2} A_c \begin{bmatrix} \frac{M(f-f_c) + M(f+f_c)}{2} \\ \frac{+j\hat{M}(f-f_c) - j\hat{M}(f+f_c)}{2} \end{bmatrix}$$



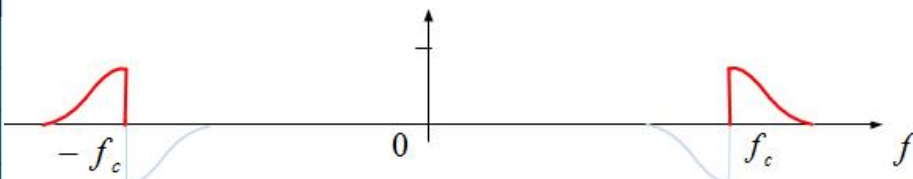
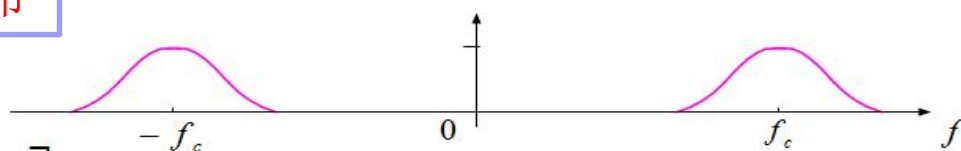
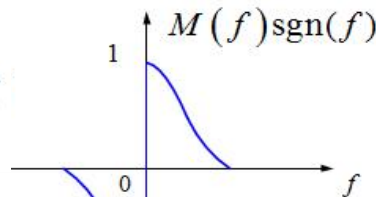
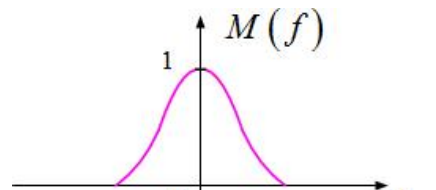
单边带调幅的相移方法

1. SSB信号公式

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t \mp \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$

$$\frac{1}{2} A_c \begin{bmatrix} m(t) \cos 2\pi f_c t \\ -\hat{m}(t) \sin 2\pi f_c t \end{bmatrix} \Leftrightarrow \begin{array}{l} -: \text{上边带} \\ +: \text{下边带} \end{array}$$

$$\frac{1}{2} A_c \begin{bmatrix} \frac{M(f-f_c) + M(f+f_c)}{2} \\ \frac{+j\hat{M}(f-f_c) - j\hat{M}(f+f_c)}{2} \end{bmatrix}$$

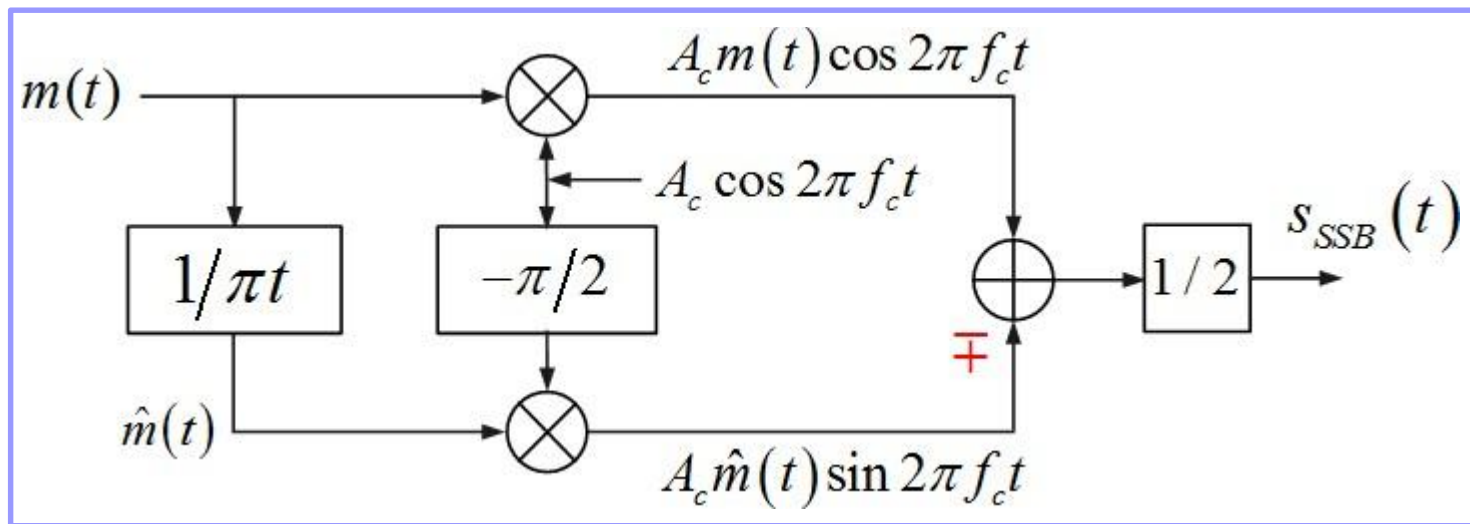


单边带调幅的相移方法

2. SSB的相移生成方法

$$s_{SSB}(t) = \frac{1}{2} A_c m(t) \cos 2\pi f_c t \mp \frac{1}{2} A_c \hat{m}(t) \sin 2\pi f_c t$$

-: 上边带
+: 下边带





第4部分 角度调制的原理

角度调制的基本原理

频带信号的一般形式

$$s(t) = A_c \cos[\underbrace{2\pi f_c t + \theta(t)}_{\text{角度}}]$$

相位

(瞬时) 角频率: $\omega_i(t) = \frac{d}{dt}[2\pi f_c t + \theta(t)] = 2\pi f_c + \frac{d}{dt}\theta(t)$

(瞬时) 频率: $f_i(t) = \frac{1}{2\pi} \omega_i(t) = \dots = f_c + \frac{1}{2\pi} \frac{d}{dt}\theta(t)$

角度调制的两种方法

FM与PM

$$s(t) = A_c \cos[2\pi f_c t + \theta(t)]$$

相位调制（调相）：

PM (Phase Modulation)

$$\theta(t) = k_{PM} m(t)$$

$$s_{PM}(t) = A_c \cos[2\pi f_c t + k_{PM} m(t)]$$

频率调制（调频）：

FM (Freq. Modulation)

$$\theta(t) = 2\pi k_{FM} \int m(t) dt$$

$$f_i(t) = f_c + k_{FM} m(t) \quad s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt\right]$$

角度调制的两种方法

FM与PM

$$s(t) = A_c \cos[2\pi f_c t + \theta(t)]$$

相位调制（调相）：
PM (Phase Modulation)

$$\theta(t) = k_{PM} m(t)$$

$$s_{PM}(t) = A_c \cos[2\pi f_c t + k_{PM} m(t)]$$

频率调制（调频）：
FM (Freq. Modulation)

$$\theta(t) = 2\pi k_{FM} \int m(t) dt$$

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt\right]$$

$$f_i(t) = f_c + k_{FM} m(t)$$

$$A_c \cos[2\pi f_c t]$$

$$f_c + k_{FM} m(t)$$

$$= A_c \cos[2\pi f_c t + 2\pi k_{FM} m(t)t]$$

$$s = vt = gt^2$$

$$gt$$

$$s = \int_0^t v(t) dt = \int_0^t gt dt = \frac{1}{2} gt^2$$

角度调制的两种方法

FM与PM

$$s(t) = A_c \cos[2\pi f_c t + \theta(t)]$$

相位调制（调相）：
PM (Phase Modulation)

$$\begin{aligned}\theta(t) &= k_{PM} m(t) \\ s_{PM}(t) &= A_c \cos[2\pi f_c t + k_{PM} m(t)]\end{aligned}$$

频率调制（调频）：
FM (Freq. Modulation)

$$\theta(t) = 2\pi k_{FM} \int m(t) dt$$

$$s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt\right]$$

$$f_i(t) = f_c + k_{FM} m(t)$$

$$\begin{aligned}A_c \cos[2\pi f_c t] \\ A_c \cos\left[2\pi \int f_i(t) dt\right]\end{aligned}$$

$$\int_0^t f_i(t) dt$$

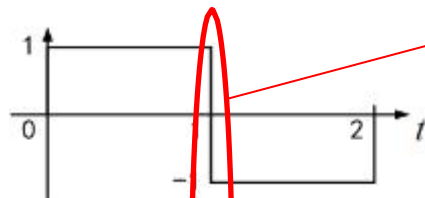
或： $\int_{-\infty}^t f_i(t) dt$

角度调制的举例

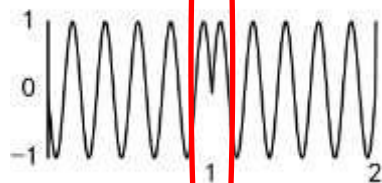
示例

- ✓ PM可能间断，带宽宽；
- ✓ FM必定连续，带宽高效

看起来，FM便于应用一些？

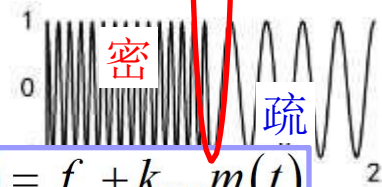


PM



$$\cos[2\pi f_c t + k_{PM} \boxed{\pm \pi} m(t)]$$

FM

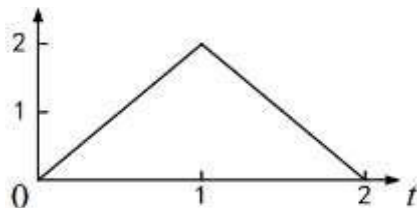
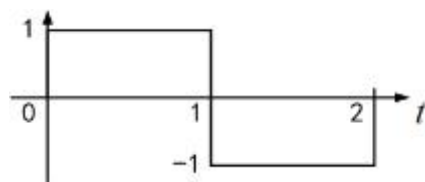


$$\cos[2\pi f_c t + \boxed{\pm 2\pi \times 10t} + C + 2\pi k_{FM} \int m(t) dt]$$

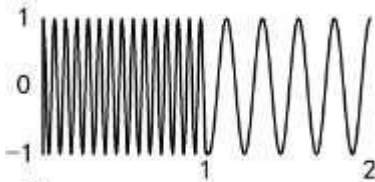
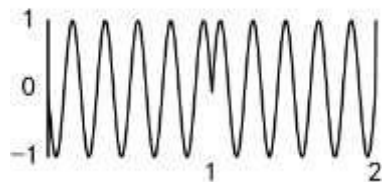
$$\boxed{f_i(t) = f_c + k_{FM} m(t)}$$

角度调制的举例

示例

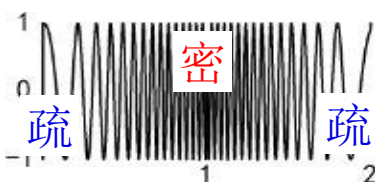
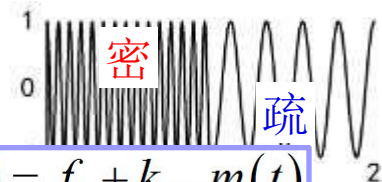


PM



$$\cos[2\pi f_c t + k_{PM} m(t)]$$

FM



$$\cos\left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt\right]$$

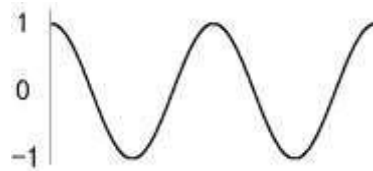
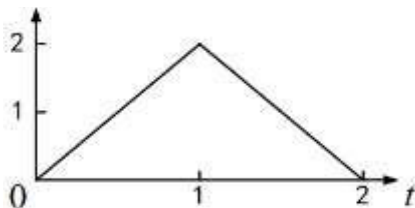
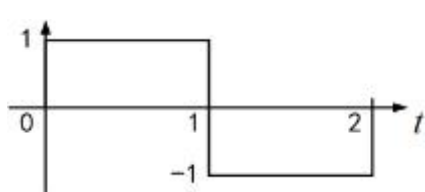
$$f_i(t) = f_c + k_{FM} m(t)$$

角度调制的举例

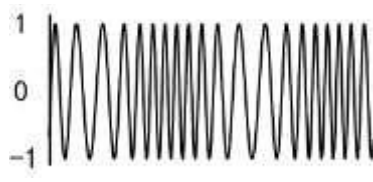
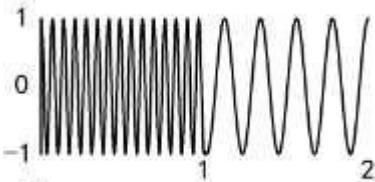
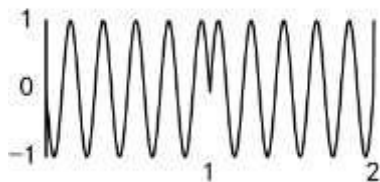
示例

同一个已调信号可以是FM与PM信号，
看起来令人困惑。

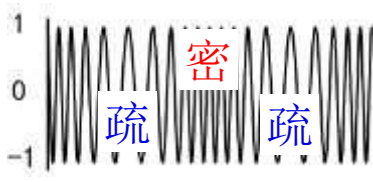
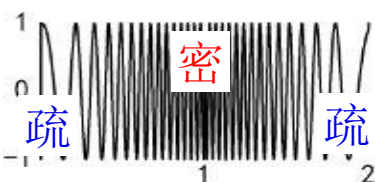
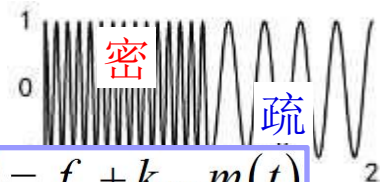
FM与PM波形是相对的、关联的。



PM



FM



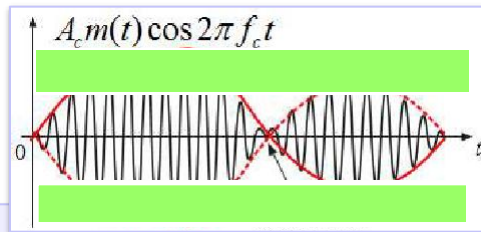
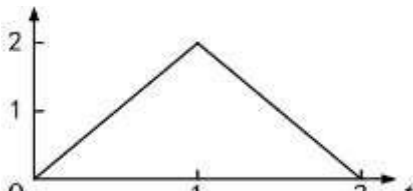
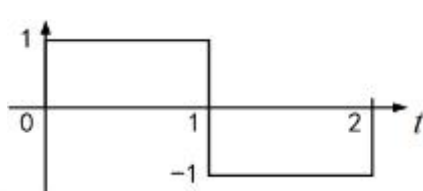
$$f_i(t) = f_c + k_{FM}m(t)$$

角度调制的举例

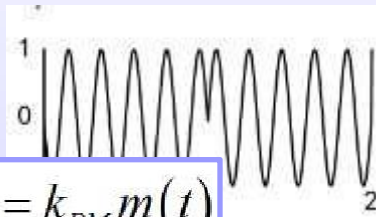
示例

FM或PM是恒幅波

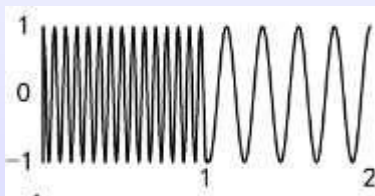
不怕非线性失真！大功率放大器的常见问题。



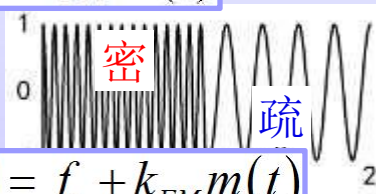
PM



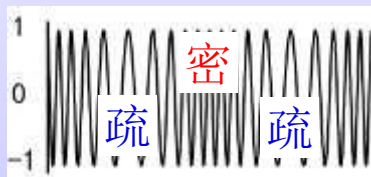
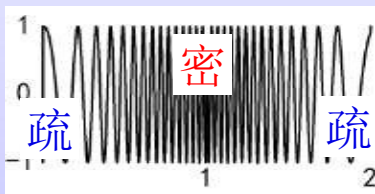
$$\theta(t) = k_{PM} m(t)$$



FM



$$f_i(t) = f_c + k_{FM} m(t)$$



角度调制的举例

示例

$$s(t) = 5 \cos \left[2\pi \times 10^6 t \pm 10 \sin 2\pi \times 10^3 t \right]$$

分析：是角度调制信号，载频1MHz。FM 或 PM？

若 $m(t) = 2 \sin 2\pi \times 10^3 t$

PM

$$\cos \left[2\pi f_c t + k_{PM} m(t) \right]$$

$$\theta(t) = 5m(t)$$

若 $m(t) = 2 \cos 2\pi \times 10^3 t$

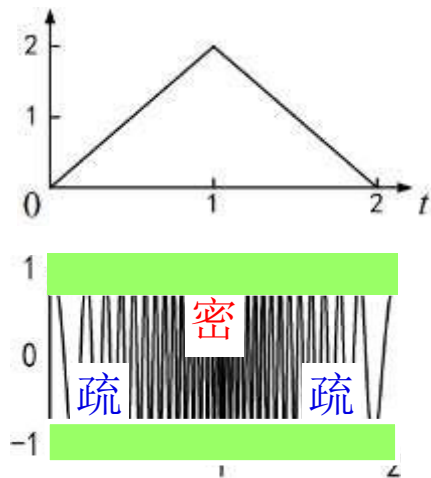
FM

$$\cos \left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt \right]$$

角度调制的基本原理

小结:

- ✓ FM与PM波形是相对的，关联的。
- ✓ FM信号平滑，更易应用



恒幅波

PM (调相) : $\theta(t) = k_{PM} m(t)$

$$s_{PM}(t) = A_c \cos \left[2\pi f_c t + \underline{k_{PM} m(t)} \right]$$

FM (调频) : $\theta(t) = 2\pi k_{FM} \int m(t) dt$ $\underline{f_i(t) = f_c + k_{FM} m(t)}$

$$s_{FM}(t) = A_c \cos \left[2\pi f_c t + \underline{2\pi k_{FM} \int m(t) dt} \right]$$

角度调制的基本参数

1. 调制的程度

最大相偏: $\Delta\theta_{\max}$

PM	$k_{PM} m(t) _{\max}$
FM	$2\pi k_{FM} \left \int m(t) dt \right _{\max}$

最大频偏: $\Delta f_{\max}$

PM	$\frac{1}{2\pi} k_{PM} m'(t) _{\max}$
FM	$k_{FM} m(t) _{\max}$

$$\text{PM} \quad \cos[2\pi f_c t + k_{PM} m(t)]$$

$$\text{FM} \quad \cos\left[2\pi f_c t + 2\pi k_{FM} \int m(t) dt\right]$$

最大频偏/相偏

- ✓ 正比于调制系数 k_{FM} 与 k_{PM}
- ✓ 正比于消息信号的（最大）幅度

角度调制的基本参数

2. 正弦消息特例与调制指数

$$m(t) = a \cos(2\pi f_m t)$$

$$\text{PM} \quad s_{PM}(t) = A_c \cos[2\pi f_c t + k_{PM} a \cos(2\pi f_m t)]$$

调相指数

$$\beta_{PM} = \Delta\theta_{\max}$$

$$\Delta\theta_{\max} = k_{PM} a$$

$$\Delta f_{\max} = f_m k_{PM} a$$

$$\text{FM} \quad s_{FM}(t) = A_c \cos\left[2\pi f_c t + \frac{k_{FM} a}{f_m} \sin(2\pi f_m t)\right]$$

调频指数

$$\beta_{FM} = \Delta f_{\max} / f_m$$

$$\Delta f_{\max} = k_{FM} a$$

角度调制的基本参数

例如 $s(t) = 5 \cos \left[2\pi \times 10^6 t \pm 10 \sin 2\pi \times 10^3 t \right]$

$$\Delta\theta_{\max} = 10 \quad \beta$$

$$\Delta f_{\max} = 10 \times 10^3$$

$$m(t) = a \cos(2\pi f_m t)$$

$$s_{PM}(t) = A_c \cos \left[2\pi f_c t + k_{PM} a \cos(2\pi f_m t) \right]$$

$$s_{FM}(t) = A_c \cos \left[2\pi f_c t + \frac{k_{FM} a}{f_m} \sin(2\pi f_m t) \right]$$

β

$$\Delta\theta_{\max} = k_{PM} a$$

$$\Delta f_{\max} = f_m k_{PM} a$$

$$\Delta f_{\max} = k_{FM} a$$

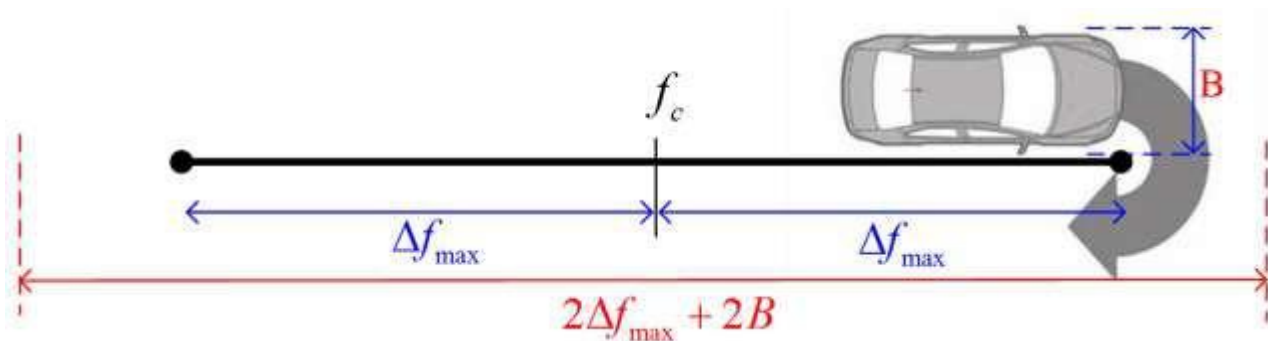
角度调制的带宽与频谱特点

1. 带宽（Carson公式）

$$D = \frac{\Delta f_{\max}}{B}$$

频偏比, 正是调制指数 β

$$B_T = 2\Delta f_{\max} + 2B = 2(D+1)B$$



角度调制的带宽与频谱特

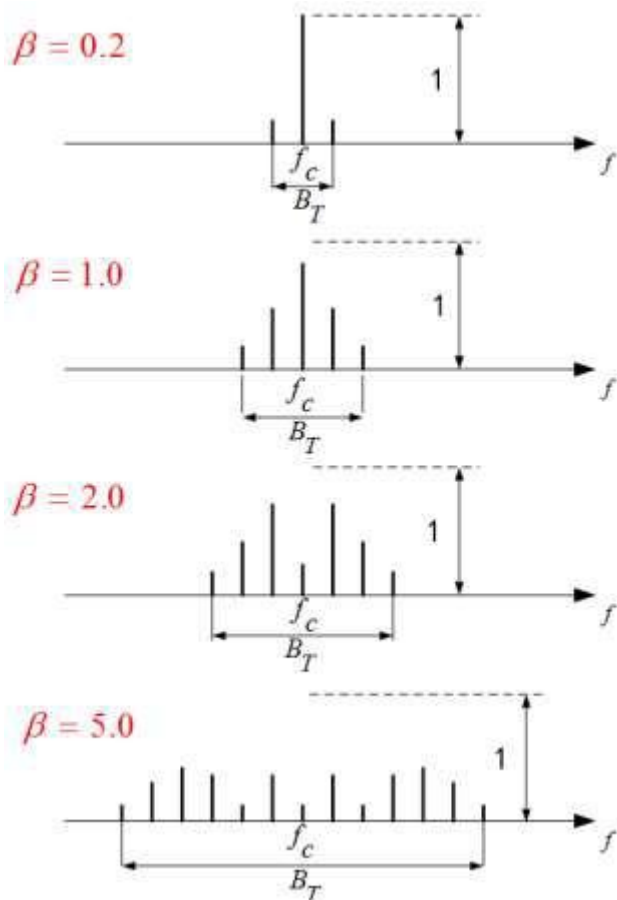
2. 频谱特点

考虑正弦消息信号的FM情况：

$$m(t) = a \cos(2\pi f_m t)$$

$$s(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]$$

- 1) $\beta \ll 1$ ：窄带角调制； β 变大时为宽带角调制；
- 2) B_T 是以 Carson 公式估算，准确地反映了信号的带宽，对应部分包括约 99% 的功率；
- 3) $m(t)$ 为正弦周期函数，产生的调制信号是周期的。
频谱形状为基频的大量谱线构成，细节变化大。



角度调制的带宽与频谱特点

例如 $s(t) = 5 \cos[2\pi \times 10^6 t \pm 10 \sin 2\pi \times 10^3 t]$

$$\Delta\theta_{\max} = 10 \quad \beta$$

$$\Delta f_{\max} = 10 \times 10^3$$

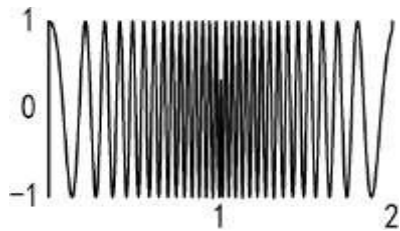
$$\boxed{B_T} = 2 \times (10 \times 10^3) + 2 \times 10^3 = \underline{22kHz}$$

$$= 2(\beta + 1)B = \underline{22kHz}$$

1kHz带宽的消息变成22kHz
BW大幅度增加

角度调制的基本参数

小结:

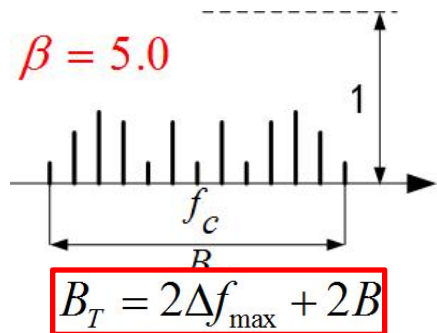


$$\text{PM} \quad \cos[2\pi f_c t + k_{PM} m(t)]$$

$$\text{FM} \quad \cos[2\pi f_c t + 2\pi k_{FM} \int m(t) dt]$$

最大相偏: $\Delta\theta_{\max}$

最大频偏: $\Delta f_{\max}$



适合信道频带: 1) 载频 f_c , 2) 带宽

- ✓ 系统核心参数: 最大频偏 $\Delta f_{\max}$ 与调制指数 $\beta_{FM} = \frac{\Delta f_{\max}}{B}$
- ✓ 电路参数 : 调频常数与消息信号的幅度

角度调制的带宽与频谱特点

案例 无线电调频 (FM) 广播与收音机

消息(音乐)带宽= 15kHz (一般语音: 3~4kHz)

$$\Delta f_{\max} = 75\text{kHz}$$

$$\beta = 75/15 = 5$$

高品质源于宽频带

- ✓ 宽带调频应用广泛
- ✓ 窄带调频应用较少

$$B_T = 2 \times 75 + 2 \times 15 = 180 \text{ (kHz)}$$

(电台频

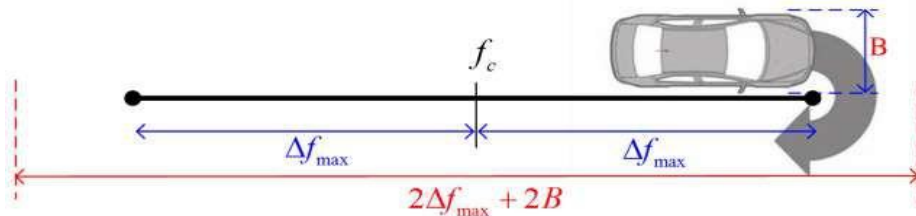
相对的, AM广播采用一般语音消息, 调制后 $2 \times 4 = 8$ (kHz)。
电台频带工作在1MHz上下(中波), 只需10kHz。干扰易引起
幅度变化, 抗干扰能力差, 音质不佳。

高品质的FM广播工作在100MHz上下(超短波), 频带需200kHz。
横幅特性利于抗干扰, 音质好, 传播距离较近。

Carson公式: $B_T = 2\Delta f_{\max} + 2B$

调频指数 β

$$\frac{\Delta f_{\max}}{B}$$



案例 无线电调频 (FM) 广播与收音机

消息(音乐)带宽= 15kHz

(一般语音: 3~4kHz)

$$\Delta f_{\max} = 75\text{kHz}$$

$$\beta = 75/15 = 5$$

$$B_T = 2 \times 75 + 2 \times 15 = 180 \text{ (kHz)} \quad (\text{电台频带} 200\text{kHz})$$

相对的, **AM广播**采用一般语音消息, 调制后 $2 \times 4 = 8$ (kHz)。电台频带工作在**1MHz**上下(中波), 只需10kHz。干扰易引起幅度变化, 抗干扰能力差, 音质不佳。

高品质的FM广播工作在**100MHz**上下(超短波), 频带需**200kHz**。横幅特性利于抗干扰, 音质好, 传播距离较近。